



Sri Lanka Technology Campus

CCS3440 - Artificial Intelligence

**Appointment No-Show Prediction Using Machine
Learning**

Technical Report

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1. INTRODUCTION

Healthcare organizations worldwide face significant challenges in resource optimization and patient care delivery. One persistent operational issue is patient no-show behavior, where individuals fail to attend scheduled medical appointments without prior cancellation. This phenomenon creates substantial inefficiencies in healthcare systems, including wasted clinical resources, increased operational costs, and reduced access to care for other patients.

The SmartCare Hospital dataset provides an opportunity to apply Artificial Intelligence (AI) and Machine Learning (ML) techniques to address this operational challenge. This project focuses on developing predictive models that can identify patients at high risk of missing appointments, enabling proactive interventions to improve attendance rates and healthcare resource utilization.

This report documents the complete AI project lifecycle, from problem definition and literature review through data preprocessing, exploratory analysis, model development, evaluation, and explanation. The final deliverable includes an AI-powered prototype that demonstrates the practical application of the developed models.

2. PROBLEM DEFINITION

2.1 Selected Problem

Option A - Appointment No-Show Prediction

2.2 Problem Statement

Many patients fail to attend scheduled medical appointments, resulting in inefficient resource utilization, increased operational costs, and lost revenue for healthcare providers. This project aims to develop an AI model capable of predicting whether a patient is likely to miss a scheduled appointment based on patient demographics, clinical measurements, historical hospital visit patterns, and financial data.

2.3 Problem Type

Binary Classification

2.4 Target Variable

no_show - This binary variable indicates whether the patient attended their scheduled appointment or failed to show up.

Output Classes:

- **No Show** (1): Patient did not attend the scheduled appointment
- **Attended** (0): Patient attended the scheduled appointment

2.5 Significance of the Problem

The problem of patient no-shows has significant implications across multiple dimensions:

Operational Impact

- Wasted clinical staff time and resources
- Inefficient utilization of medical facilities
- Disruption to appointment scheduling systems
- Increased wait times for other patients

Financial Impact

- Lost revenue from missed appointments
- Increased operational costs due to inefficiency

- Reduced capacity to serve patients
- Negative impact on healthcare organization profitability

Patient Care Impact

- Delayed treatment for patients who miss appointments
- Potential deterioration of health conditions
- Reduced continuity of care
- Lower patient satisfaction scores

3. LITERATURE REVIEW

3.1 Overview

Appointment no-show prediction using machine learning has been an active area of healthcare research over the past decade. A review of five recent peer-reviewed studies reveals consistent patterns in the algorithms used, the predictors identified, and the challenges that remain unresolved in translating predictive models into real-world clinical impact. Table I summarizes the key characteristics of each reviewed paper.

Title	Dataset	Algorithms Used	Key Result	Limitations
Predicting Patient No-Shows Using Machine Learning: A Comprehensive Review and Future Research Agenda (2025) [1]	52 studies (2010-2025), review paper	Logistic Regression, Random Forest, XGBoost, Decision Trees, SVM, Deep Learning	AUC range of 0.75-0.95 across reviewed studies; Logistic Regression most frequently used	Limited model interpretability, poor healthcare system integration, data quality issues
Predicting Missed Appointments in Primary Care: A Personalized Machine Learning Approach (2025) [2]	109,328 patients; 1.1M+ primary care appointments	Gradient Boosting, Random Forest, Neural Networks, Logistic Regression	Gradient Boosting best (AUROC = 0.852); SHAP used for interpretability	Complex feature integration (geosocial/environmental data may not generalize)

Toffaha et al. (2024) - Predicting Hospital No-Shows: Interpretable ML Models Approach [3]	Brazilian Medical Appointment No-Shows dataset	12 classification algorithms incl. Tree-GAM, FIGS, TAO Tree	Interpretable models achieved strong accuracy with transparent reasoning	Advanced interpretable models had longer computation time
Dunstan et al. (2023) - Predicting No-Show Appointments in a Pediatric Hospital in Chile [4]	395,963 appointments (2015-2018)	Logistic Regression, Random Forest, Balanced Random Forest, Balanced Bagging, AdaBoost, RUSBoost, Easy Ensemble, SVM	Balanced Bagging & Balanced Random Forest best; real-world trial reduced no-shows from 21.0% to 10.7%	Class imbalance required specialized ensemble techniques
No-Show in Medical Appointments with Machine Learning Techniques: A Systematic Literature Review (2017-2022) [5]	24 studies reviewed (SLR, PRISMA)	Decision Trees, Logistic Regression, Random Forest, Gradient Boosting, Neural Networks	Gradient Boosting, Random Forest, Decision Trees performed best; one study showed real-world reduction	Most studies model-only, not deployed; low real-world adoption

			(19.3% - 15.9%)	
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Table 1: summary of reviewed literature

3.2 Existing AI Approaches and Algorithms

Across the literature, a wide range of machine learning algorithms have been applied to no-show prediction. Toffaha and Simsekler [1] reviewed 52 studies published between 2010 and 2025 and found that Logistic Regression was the most commonly used algorithm, appearing in 68% of the studies reviewed, while tree-based models, ensemble methods, and deep learning techniques have gained traction more recently. This trend toward ensemble and boosting-based methods is echoed in Tuan et al. [2], who compared Gradient Boosting, Random Forest, Neural Networks, and LASSO Logistic Regression on over one million primary care appointments, finding that Gradient Boosting achieved the strongest result with an AUROC of 0.852 for no-show prediction. Similarly, Dunstan et al. [4] tested a broad set of algorithms-including Logistic Regression, Random Forest, Balanced Random Forest, Balanced Bagging, AdaBoost, RUSBoost, Easy Ensemble, and SVM-on nearly 400,000 appointments from a Chilean pediatric hospital, finding that Balanced Bagging and Balanced Random Forest performed best, particularly in handling the class imbalance typical of no-show datasets.

A separate line of research has prioritized model interpretability over raw predictive power. Toffaha et al. [3] tested twelve classification algorithms on the Brazilian Medical Appointment No-Shows dataset, focusing specifically on interpretable model types such as Tree-GAM, FIGS, and TAO Tree, and found that these models could achieve strong accuracy while still producing transparent, human-understandable reasoning-a critical requirement in clinical settings where practitioners must understand and trust model outputs.

Salazar et al. [5], in a systematic literature review covering studies from 2017–2022, corroborate these findings at a broader scale, reporting that decision-tree-based algorithms were generally the most appropriate for building no-show prediction models. The same review noted that while neural networks showed predictive promise, they were limited by poor interpretability in clinical contexts-reinforcing the pattern seen in Toffaha et al. [3].

Taken together, these findings support the algorithm choices made in this project: Logistic Regression, Decision Tree, and Random Forest are not only among the most widely used algorithms in the domain, but also represent a useful spectrum from highly interpretable

(Logistic Regression, Decision Tree) to higher-performing ensemble methods (Random Forest).

3.3 Common Predictors of No-Shows

Despite differences in dataset size and geography, the literature converges on a consistent set of predictive features. Prior no-show history, scheduling lead time, and patient age were repeatedly identified as the strongest predictors of missed appointments across studies. Tuan et al. [2] additionally found that lead time-the number of days between booking and visit-was the single strongest predictor, with lead times longer than 60 days associated with a greater risk of missed appointments, while provider experience and socioeconomic factors also contributed meaningfully. Dunstan et al. [4] confirmed that historical no-show behavior and reservation delay were the two strongest predictors in their dataset. Salazar et al. [5] further identified secondary factors such as clinic distance, medical specialty, and appointment confirmation status as relevant, though less dominant, predictors.

3.4 Real-World Impact and Interventions

While most published research focuses on model development rather than deployment, a small number of studies have measured real-world outcomes. Dunstan et al. [4] is notable in this regard: an 8-week trial using model-driven targeted reminder calls reduced the no-show rate from 21.0% to 10.7%. Similarly, Salazar et al. [5] identified one study within their review (Chong et al.) that achieved a real-world reduction from 19.3% to 15.9% using targeted reminders informed by predictive modeling. These findings demonstrate that predictive models, when paired with operational interventions such as reminder calls, overbooking, or rescheduling prompts, can produce measurable reductions in no-show rates.

3.5 Explainability and Fairness

Explainable AI (XAI) has emerged as a growing focus in this domain, driven by the need for transparency in healthcare decision-making. Tuan et al. [2] applied SHAP (Shapley Additive Explanations) to identify influential features and additionally conducted a subgroup fairness analysis across sex and race/ethnicity, confirming that model performance remained consistent across demographic groups. Toffaha et al. [3] similarly emphasized that advanced interpretable models can offer both accuracy and transparency, though at the cost of longer computation times, which may limit their practicality in time-sensitive clinical environments.

3.6 Research Gap Analysis

Based on the reviewed literature, the following research gaps we were identified:

1. **Limited interpretability in high-performing models:** While ensemble and boosting methods (Gradient Boosting, XGBoost) often achieve the best predictive performance, they are frequently less interpretable than simpler models-creating tension between accuracy and clinical trust [1], [3].
2. **Insufficient integration with real healthcare systems:** Most studies stop at model development and evaluation, with very few demonstrating real-world deployment or measuring operational impact [5].
3. **Data quality and class imbalance challenges:** No-show datasets are typically imbalanced, requiring techniques such as SMOTE or balanced ensemble methods to address effectively [4].
4. **Lack of standardized evaluation frameworks:** Studies vary widely in the metrics, datasets, and validation approaches used, making it difficult to compare model performance across the literature [1].
5. **Underuse of Explainable AI:** Although XAI techniques such as SHAP are increasingly applied, they remain the exception rather than the norm, particularly in smaller or less-resourced studies [2].

4. DATASET DESCRIPTION

4.1 Dataset Overview

The SmartCare Hospital AI Dataset contains 1000 patient records specifically designed for Artificial Intelligence and Machine Learning applications in healthcare. The dataset was provided as part of the coursework and includes comprehensive information about patients, their clinical characteristics, hospital operations data, and financial information.

Dataset Details:

- File Name: smartcare_ai_dataset_1000.csv
- Total Records: 1000
- Total Features: 33
- No Missing Values
- Target Variable: no_show (Appointment No-Show Prediction)

4.2 Dataset Loading and Initial Inspection

4.2.1 Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)

print("Libraries imported successfully.")

Libraries imported successfully.
```

Figure 1: Import statements showing pandas, numpy, matplotlib, seaborn

The required libraries (pandas for data manipulation, numpy for numerical operations, matplotlib and seaborn for visualization) are imported to enable dataset exploration and analysis.

4.2.2 Load Dataset

```
df = pd.read_csv("smartcare_ai_dataset_1000.csv")

print("Dataset loaded successfully.")

Dataset loaded successfully.
```

Figure 2: Dataset loading code and confirmation message

The dataset was successfully loaded from the CSV file into a pandas DataFrame, confirming the file path and data accessibility.

4.2.3: Display First Five Records

```
df.head()
```

	record_id	patient_id	age	gender	blood_group	department	diagnosis	appointment_date	waiting_days	previous_appointments	missed_previous_appointments	appointment_status	admitted
0	1	P10001	23	Male	A-	General Medicine	Migraine	2025-04-18	10	1	0	Completed	0
1	2	P10002	28	Male	B-	General Medicine	Diabetes	2025-05-15	2	3	1	Completed	0
2	3	P10003	22	Male	B+	Orthopedics	Back Pain	2025-07-08	22	7	1	No-Show	0
3	4	P10004	46	Female	AB-	Cardiology	Asthma	2025-10-18	16	1	0	Completed	0
4	5	P10005	51	Female	O+	Neurology	Hypertension	2025-13-18	12	4	0	Scheduled	0

Figure 3: First 5 rows of the dataset

The first five records display the structure of the dataset, showing patient identifiers (patient_id), demographics (age, gender, blood_group), clinical information (diagnosis, blood pressure components, blood sugar, cholesterol, BMI), hospital operations data (department, appointment details, admission status), and financial information (consultation fees, room charges, lab charges, medicine charges, total bill).

4.2.4: Display Last Five Records

	record_id	patient_id	age	gender	blood_group	department	diagnosis	appointment_date	waiting_days	previous_appointments
996	996	P10996	54	Male	AB+	Laboratory Services	Fever	2025-03-24	24	1
998	997	P10997	53	Male	O-	Orthopedics	Fracture	2025-05-20	35	4
997	998	P10998	26	Male	A+	Cardiology	Chest Pain	2025-02-07	9	3
998	999	P10999	41	Female	B-	Cardiology	Fracture	2025-07-10	27	1
999	1000	P11000	22	Female	A-	Radiology	Chest Pain	2025-01-08	17	3

Figure 4: Last 5 rows of the dataset

The last five records confirm the dataset structure is consistent throughout all 1000 rows, with no unexpected changes in data format or type at the end of the file.

4.2.5: Dataset Dimensions

```
Number of Records: 1000
Number of Attributes: 33
Dataset Shape: (1000, 33)
```

Figure 5: Output showing (1000, 33)

The dataset contains **1,000 rows** (patient records) and **33 columns** (attributes), providing a rich set of features for predicting appointment no-show behavior.

4.3 Attribute Description and Data Types

4.3.1 List All Attributes

```
Total number of attributes: 33

Attribute Names:

01. record_id
02. patient_id
03. age
04. gender
05. blood_group
06. department
07. diagnosis
08. appointment_date
09. waiting_days
10. previous_appointments
11. missed_previous_appointments
12. appointment_status
13. admitted
14. room_type
15. length_of_stay_days
16. previous_admissions
17. systolic_bp
18. diastolic_bp
19. blood_sugar_mg_dl
20. cholesterol_mg_dl
21. bmi
22. lab_tests_count
23. treatments_count
24. consultation_fee_lkr
25. room_charge_lkr
26. lab_charge_lkr
27. medicine_charge_lkr
28. total_bill_lkr
29. payment_status
30. payment_method
31. no_show
32. readmitted_30_days
33. disease_risk_level
```

Figure 6: Full list of 33 column names

The dataset contains 33 attributes including:

- Identifiers: record_id, patient_id
- Demographics: age, gender, blood_group
- Clinical: diagnosis, blood pressure (systolic/diastolic), blood sugar, cholesterol, bmi
- Appointment: appointment_date, waiting_days, appointment_status
- History: previous_appointments, missed_previous_appointments, previous_admissions, length_of_stay_days
- Services: admitted, room_type, lab_tests_count, treatments_count
- Financial: consultation_fee_lkr, room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr, payment_status, payment_method
- Targets: no_show, readmitted_30_days, disease_risk_level

4.3.2 Dataset Information

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 33 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   record_id                            1000 non-null   int64  
 1   patient_id                           1000 non-null   object  
 2   age                                  1000 non-null   int64  
 3   gender                               1000 non-null   object  
 4   blood_group                          1000 non-null   object  
 5   department                           1000 non-null   object  
 6   diagnosis                            1000 non-null   object  
 7   appointment_date                     1000 non-null   object  
 8   waiting_days                         1000 non-null   int64  
 9   previous_appointments                1000 non-null   int64  
10   missed_previous_appointments         1000 non-null   int64  
11   appointment_status                   1000 non-null   object  
12   admitted                             1000 non-null   int64  
13   room_type                            94 non-null     object  
14   length_of_stay_days                  1000 non-null   int64  
15   previous_admissions                  1000 non-null   int64  
16   systolic_bp                         1000 non-null   int64  
17   diastolic_bp                        1000 non-null   int64  
18   blood_sugar_mg_dl                   1000 non-null   int64  
19   cholesterol_mg_dl                   1000 non-null   int64  
20   bmi                                  1000 non-null   float64 
21   lab_tests_count                      1000 non-null   int64  
22   treatments_count                     1000 non-null   int64  
23   consultation_fee_lkr                 1000 non-null   int64  
24   room_charge_lkr                     1000 non-null   int64  
25   lab_charge_lkr                       1000 non-null   int64  
26   medicine_charge_lkr                  1000 non-null   int64  
27   total_bill_lkr                       1000 non-null   int64  
28   payment_status                       1000 non-null   object  
29   payment_method                       1000 non-null   object  
30   no_show                             1000 non-null   int64  
31   readmitted_30_days                   1000 non-null   int64  
32   disease_risk_level                   1000 non-null   object  
dtypes: float64(1), int64(21), object(11)
memory usage: 257.9+ KB
```

Figure 7: Complete `df.info()` output showing column names, non-null counts, and data types

The dataset info confirms:

- 1000 entries with no missing values across all 33 columns
- 21 integer (int64) columns: record_id, age, waiting_days, previous_appointments, missed_previous_appointments, admitted, length_of_stay_days, previous_admissions, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, lab_tests_count, treatments_count, consultation_fee_lkr, room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr, no_show, readmitted_30_days
- 11 object (categorical) columns: patient_id, gender, blood_group, department, diagnosis, appointment_date, appointment_status, room_type, payment_status, payment_method, disease_risk_level
- 1 float (float64) column: bmi
- Memory usage: 257.9+ KB

4.3.3 Data Type Summary

```
Data Type Summary:
int64      21
object     11
float64     1
Name: count, dtype: int64
```

Figure 8: Data type count summary

The dataset has 21 integer attributes (64%), 11 object/categorical attributes (33%), and 1 float attribute (3%). This mixed data type structure requires different preprocessing approaches: numerical features will need scaling, while categorical features will require encoding.

4.3.4 Numerical Attributes

```
Number of numerical attributes: 22

Numerical Attributes:
- record_id
- age
- waiting_days
- previous_appointments
- missed_previous_appointments
- admitted
- length_of_stay_days
- previous_admissions
- systolic_bp
- diastolic_bp
- blood_sugar_mg_dl
- cholesterol_mg_dl
- bmi
- lab_tests_count
- treatments_count
- consultation_fee_lkr
- room_charge_lkr
- lab_charge_lkr
- medicine_charge_lkr
- total_bill_lkr
- no_show
- readmitted_30_days
```

Figure 9: List of 22 numerical attributes

The numerical attributes include:

- Identifiers/Indices: record_id
- Demographic: age
- Clinical: systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi
- Appointment: waiting_days
- History: previous_appointments, missed_previous_appointments, previous_admissions, length_of_stay_days
- Services: admitted, lab_tests_count, treatments_count
- Financial: consultation_fee_lkr, room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr
- Targets: no_show, readmitted_30_days

4.3.5 Categorical Attributes

```
Number of categorical attributes: 11

Categorical Attributes:
- patient_id
- gender
- blood_group
- department
- diagnosis
- appointment_date
- appointment_status
- room_type
- payment_status
- payment_method
- disease_risk_level
```

Figure 10: List of 11 categorical attributes

The categorical attributes include:

- Identifiers: patient_id
- Demographic: gender, blood_group
- Clinical: diagnosis
- Operational: department, appointment_date, appointment_status, room_type
- Financial: payment_status, payment_method
- Alternative Target: disease_risk_level

4.4 Statistical Summary

	count	mean	std	min	25%	50%	75%	max
record_id	1000.0	500.5000	288.819436	1.0	250.75	500.5	750.25	1000.0
age	1000.0	44.7440	17.851474	1.0	33.00	44.0	57.00	90.0
waiting_days	1000.0	21.8510	13.040892	0.0	11.00	22.0	34.00	44.0
previous_appointments	1000.0	2.8840	1.693343	0.0	2.00	3.0	4.00	10.0
missed_previous_appointments	1000.0	0.5530	0.740094	0.0	0.00	0.0	1.00	4.0
admitted	1000.0	0.3300	0.470448	0.0	0.00	0.0	1.00	1.0
length_of_stay_days	1000.0	1.1030	1.885191	0.0	0.00	0.0	2.00	9.0
previous_admissions	1000.0	0.8550	0.958590	0.0	0.00	1.0	1.00	5.0
systolic_bp	1000.0	128.4230	15.486895	85.0	117.00	128.0	139.00	178.0
diastolic_bp	1000.0	78.8280	10.043669	50.0	72.00	79.0	86.00	111.0
blood_sugar_mg_dl	1000.0	114.8320	26.017007	65.0	97.00	114.0	132.00	201.0
cholesterol_mg_dl	1000.0	204.8140	36.645577	100.0	180.00	204.0	230.00	330.0
bmi	1000.0	25.5191	4.252852	14.0	22.70	25.6	28.30	38.8
lab_tests_count	1000.0	2.2920	1.679749	0.0	1.00	2.0	3.00	10.0
treatments_count	1000.0	1.8050	1.596723	0.0	1.00	1.0	3.00	10.0
consultation_fee_ikr	1000.0	2820.5000	837.694146	1500.0	2000.00	3000.0	3500.00	4000.0
room_charge_ikr	1000.0	2412.5000	11140.880260	0.0	0.00	0.0	0.00	150000.0
lab_charge_ikr	1000.0	4869.6000	4122.515127	0.0	2400.00	3600.0	6300.00	27000.0
medicine_charge_ikr	1000.0	7971.0000	4629.777041	506.0	4316.50	7680.0	11032.00	20371.0
total_bill_ikr	1000.0	18073.6000	13710.258789	2954.0	11018.00	15068.5	20870.00	179528.0
no_show	1000.0	0.4950	0.500225	0.0	0.00	0.0	1.00	1.0
readmitted_30_days	1000.0	0.2530	0.434948	0.0	0.00	0.0	1.00	1.0

Figure 11: Complete df.describe() output showing count, mean, std, min, quartiles, and max for all numerical features

The statistical summary provides key insights into the distribution of numerical features:

- Age: Ranges from 1-90 years with a mean of 44.74 years, indicating a diverse patient population including pediatric and elderly patients
- Waiting Days: Ranges from 0-44 days with a mean of 21.85 days, showing considerable variation in appointment scheduling lead times
- Previous Appointments: Most patients have 2-4 previous appointments (mean 2.88), with a maximum of 10
- Blood Pressure: Systolic BP ranges 85-178 (mean 128.42) and diastolic BP ranges 50-111 (mean 78.83), suggesting a mix of normal and hypertensive patients
- Blood Sugar: Ranges 65-201 mg/dL (mean 114.83), indicating a mix of normal, pre-diabetic, and diabetic patients
- BMI: Ranges 14.0-38.8 (mean 25.52), showing patients across underweight to obese categories
- Total Bill: Ranges from 2,954-179,528 LKR (mean 18,073.60), reflecting the diversity of patient care needs and costs

4.5 Categorical Attribute Summary

<pre> Attribute: patient_id Number of unique values: 1800 Value counts: patient_id P1000 1 P10001 1 P10002 1 P10003 1 P10004 1 .. P10013 1 P10012 1 P10011 1 P10010 1 P10009 1 Name: count, Length: 1800, dtype: int64 Attribute: gender Number of unique values: 2 Value counts: gender Male 501 Female 499 Name: count, dtype: int64 Attribute: blood_group Number of unique values: 8 Value counts: blood_group AB- 136 A- 135 O- 135 B+ 132 AB+ 122 A+ 120 B+ 112 O+ 108 Name: count, dtype: int64 Attribute: department Number of unique values: 7 Value counts: department General Medicine 159 Cardiology 158 Neurology 148 Orthopedics 141 Radiology 140 Pediatrics 138 Laboratory Services 116 </pre>	<pre> Pediatrics 138 Laboratory Services 116 Name: count, dtype: int64 Attribute: diagnosis Number of unique values: 10 Value counts: diagnosis Pneumonia 115 Fever 107 Back Pain 104 Migraine 104 Fracture 104 Kidney Infection 101 Diabetes 97 Asthma 92 Hypertension 89 Chest Pain 87 Name: count, dtype: int64 Attribute: appointment_date Number of unique values: 339 Value counts: appointment_date 2025-08-08 8 2025-12-17 7 2025-01-05 7 2025-07-20 7 2025-04-01 7 .. 2025-01-31 1 2025-08-12 1 2025-08-31 1 2025-11-21 1 2025-07-14 1 Name: count, Length: 339, dtype: int64 Attribute: appointment_status Number of unique values: 4 Value counts: appointment_status No-Show 495 Completed 388 Scheduled 72 Cancelled 53 Name: count, dtype: int64 Attribute: room_type Number of unique values: 3 Value counts: room_type New 506 General Ward 52 Private Room 33 ICU 9 Name: count, dtype: int64 </pre>	<pre> Attribute: room_type Number of unique values: 3 Value counts: room_type New 506 General Ward 52 Private Room 33 ICU 9 Name: count, dtype: int64 Attribute: payment_status Number of unique values: 3 Value counts: payment_status Paid 668 Unpaid 212 Partially Paid 120 Name: count, dtype: int64 Attribute: payment_method Number of unique values: 4 Value counts: payment_method Insurance 259 Cash 257 Online 244 Card 240 Name: count, dtype: int64 Attribute: disease_risk_level Number of unique values: 3 Value counts: disease_risk_level Medium 469 High 400 Low 131 Name: count, dtype: int64 </pre>
---	---	---

Figure 12: Categorical Attribute Value Counts

The categorical value counts reveal important distribution patterns:

Gender:

- 501 Male (50.1%) and 499 Female (49.9%), providing balanced gender representation.

Blood Group:

- AB- is most common (136), followed by A- (135) and O- (135)
- O+ is least common (108)
- Distribution is relatively even across all eight blood types.

Department:

- General Medicine (159) and Cardiology (158) are the most represented
- Neurology (148), Orthopedics (141), and Radiology (140) moderate
- Pediatrics (138) and Laboratory Services (116) have fewer patients

Diagnosis:

- Pneumonia (115) is most common, followed by Fever (107)
- Back Pain (104), Migraine (104), and Fracture (104) equally represented
- Chest Pain (87) is least common

Appointment Status:

- No-Show (495) and Completed (380) are most common
- Scheduled (72) and Cancelled (53) represent smaller portions

Payment Status:

- Paid (668) is the majority (66.8%)
- Unpaid (212) and Partially Paid (120) represent 33.2%

Payment Method:

- Insurance (259), Cash (257), Online (244), and Card (240) - relatively balanced

Disease Risk Level:

- Medium (469) is most common, followed by High (400) and Low (131)

4.6 Input and Target Variable Identification

4.6.1 Identify Input and Target Variables

```
Target Variable:
no_show

Number of Input Variables: 32

Input Variables:
- record_id
- patient_id
- age
- gender
- blood_group
- department
- diagnosis
- appointment_date
- waiting_days
- previous_appointments
- missed_previous_appointments
- appointment_status
- admitted
- room_type
- length_of_stay_days
- previous_admissions
- systolic_bp
- diastolic_bp
- blood_sugar_mg_dl
- cholesterol_mg_dl
- bmi
- lab_tests_count
- treatments_count
- consultation_fee_lkr
- room_charge_lkr
- lab_charge_lkr
- medicine_charge_lkr
- total_bill_lkr
- payment_status
- payment_method
- readmitted_30_days
- disease_risk_level
```

Figure 13: Target variable identified as "no_show" with list of 32 input variables

The target variable "no_show" is clearly identified, with all remaining 32 columns serving as input features. This includes patient demographics, clinical measurements, hospital operations data, and financial information - providing a comprehensive feature set for predictive modeling.

4.6.2 Verify Target Variable

```
Target Variable: no_show
Data Type: int64
Number of Unique Values: 2

Unique Values:
[0 1]

Target Value Counts:
no_show
0      505
1      495
Name: count, dtype: int64
```

Figure 14: Target variable verification

The target variable "no_show" is confirmed as:

- Data Type: Integer (int64)
- Unique Values: 0 (Attended) and 1 (No-Show)
- Distribution: 505 Attended (50.5%), 495 No-Show (49.5%)

This confirms the dataset is configured for binary classification.

4.6.3 Target Class Distribution

```
Target Class Distribution:
no_show
0      505
1      495
Name: count, dtype: int64

Target Class Percentages:
no_show
0      50.5
1      49.5
Name: count, dtype: float64
```

Figure 15: Target class distribution with counts and percentages

The dataset shows a well-balanced distribution:

- Attended (0): 505 patients (50.5%)
- No-Show (1): 495 patients (49.5%)

This near-perfect balance is favorable for model training without requiring extensive resampling techniques.

4.6.4 Target Distribution Visualization

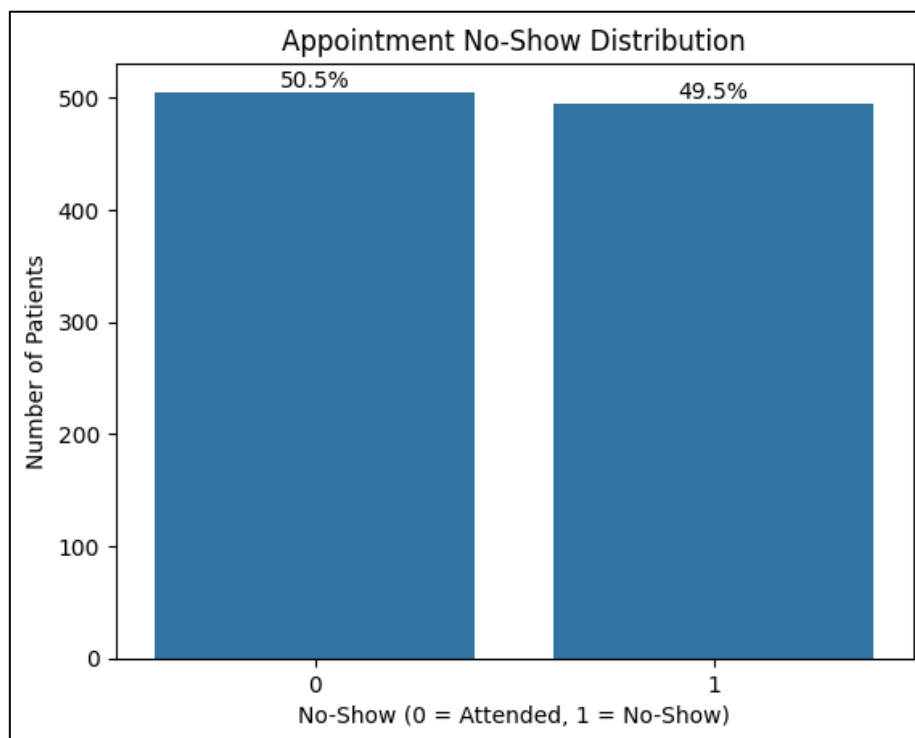


Figure 16: Bar chart showing target distribution with values displayed on bars

The bar chart visually confirms the balanced nature of the target variable. The 50.5% / 49.5% split means the dataset does not suffer from significant class imbalance, which is advantageous for machine learning models as most algorithms perform optimally with balanced datasets.

4.7 Data Quality Assessment

4.7.1 Missing Values Analysis

	Missing values	Missing Percentage
room type	906	90.6
record id	0	0.0
patient id	0	0.0
gender	0	0.0
age	0	0.0
department	0	0.0
diagnosis	0	0.0
appointment date	0	0.0
blood group	0	0.0
waiting days	0	0.0
previous appointments	0	0.0
appointment status	0	0.0
mixed previous appointments	0	0.0
admitted	0	0.0
length of stay days	0	0.0
previous admissions	0	0.0
systolic bp	0	0.0
diastolic bp	0	0.0
blood sugar mg dl	0	0.0
cholesterol mg dl	0	0.0
hba1c	0	0.0
lab test count	0	0.0
treatment count	0	0.0
consultation fee lir	0	0.0
room charge lir	0	0.0
lab charge lir	0	0.0
medicine charge lir	0	0.0
total bill lir	0	0.0
payment status	0	0.0
payment method	0	0.0
no show	0	0.0

Figure 17: Missing values summary

The missing value analysis reveals:

- Only room_type contains missing values (906 out of 1000 records)
- All other 32 columns have zero missing values
- room_type is only applicable to patients who were admitted to the hospital, explaining the missing values for non-admitted patients

This indicates excellent data completeness and reduces the need for imputation strategies during preprocessing.

4.7.2 Duplicate Record Analysis

```
Number of duplicate records: 0
```

Figure 18: Duplicate count

No duplicate rows exist in the dataset. This ensures data integrity and eliminates the need for duplicate removal during preprocessing, preserving all 1000 unique patient records for analysis.

4.7.3 Unique Values / Cardinality

unique_summary			
	Attribute	Data Type	Unique Values
0	record_id	int64	1000
1	patient_id	object	1000
2	age	int64	88
3	gender	object	2
4	blood_group	object	8
6	department	object	7
8	diagnosis	object	10
7	appointment_date	object	339
8	waiting_days	int64	45
9	previous_appointments	int64	10
10	missed_previous_appointments	int64	5
11	appointment_status	object	4
12	admitted	int64	2
13	room_type	object	3
14	length_of_stay_days	int64	10
16	previous_admissions	int64	6
18	systolic_bp	int64	83
17	diastolic_bp	int64	58
18	blood_sugar_mg_dl	int64	120
19	cholesterol_mg_dl	int64	172
20	bmi	float64	197
21	lab_tests_count	int64	11
22	treatments_count	int64	11
23	consultation_fee_lkr	int64	6
24	room_charge_lkr	int64	22
26	lab_charge_lkr	int64	29
28	medicine_charge_lkr	int64	966
27	total_bill_lkr	int64	981
28	payment_status	object	3
29	payment_method	object	4
30	no_show	int64	2

Figure 19: *unique_summary DataFrame with columns*

The cardinality analysis reveals that most categorical features have low cardinality (2-10 unique values), making them suitable for encoding without dimensionality issues. The `patient_id` (1000 unique) and `appointment_date` (339 unique) have high cardinality and require careful handling. Numerical features like `cholesterol` (169 unique), `bmi` (193 unique), and `total_bill_lkr` (197 unique) show significant variation, providing rich information for predictive modeling.

4.7.4 Check Missing Target Values

```
Missing target values: 0
Target values:
no_show
0      585
1      495
Name: count, dtype: int64
```

Figure 20: *Missing Target Values Check*

The target variable "no_show" has 0 missing values with a balanced distribution of 505 Attended (50.5%) and 495 No-Show (49.5%). This complete, balanced target data eliminates the need for imputation or data removal and provides an ideal foundation for binary classification modeling.

4.7.5 Final Dataset Overview

```
=====
TASK 02 - DATASET UNDERSTANDING SUMMARY
=====
Dataset Size      : (1000, 33)
Total Records     : 1000
Total Attributes  : 33
Numerical Features : 22
Categorical Features : 11
Missing Values    : 906
Duplicate Records  : 0
Target Variable   : no_show
Missing Target    : 0

Task 02 completed successfully.
```

Figure 21: Final summary

The final overview validates that the dataset is:

- Complete: Only room_type has missing values (expected for non-admitted patients)
- Unique: Zero duplicates
- Well-structured: Mix of 22 numerical and 11 categorical features
- Ready for preprocessing: All necessary information is available for the next stages

4.8 Data Quality Implications

Based on the step-by-step analysis above, the following data quality implications we were identified:

1. **Completeness Advantage:** The absence of missing values (except room_type which is naturally missing for non-admitted patients) simplifies preprocessing and eliminates the need for imputation strategies.
2. **No Duplicates:** 0 duplicate records ensure data integrity and preserve all 1000 unique patient records.
3. **Class Balance:** The near-balanced target distribution (50.5% Attended vs 49.5% No-Show) is favorable for model training without extensive resampling.
4. **Feature Diversity:** The combination of 22 numerical and 11 categorical features provides comprehensive information for predictive modeling.

5. Clinical Validity: The ranges of clinical measurements appear clinically plausible, supporting the dataset's utility for healthcare AI applications.
6. Categorical Variety: The 11 categorical features with balanced distributions provide meaningful groupings for predictive analysis.

5. DATA PREPROCESSING

5.1 Initial Data Quality Check

Rows: 1000 Columns: 33		
Data types:		
	Data Type	
record_id	Int64	
patient_id	object	
age	Int64	
gender	object	
blood_group	object	
department	object	
diagnosis	object	
appointment_date	object	
waiting_days	Int64	
previous_appointments	Int64	
missed_previous_appointments	Int64	
appointment_status	object	
admitted	Int64	
room_type	object	
length_of_stay_days	Int64	
previous_admissions	Int64	
systolic_bp	Int64	
diastolic_bp	Int64	
blood_sugar_mg_dl	Int64	
cholesterol_mg_dl	Int64	
bmi	float64	
lab_tests_count	Int64	
treatments_count	Int64	
consultation_fee_lkr	Int64	
room_charge_lkr	Int64	
lab_charge_lkr	Int64	
medicines_charge_lkr	Int64	
total_bill_lkr	Int64	
payment_status	object	
payment_method	object	
no_show	Int64	
readmitted_30_days	Int64	
disease_risk_level	object	
Missing values:	Missing Values	
record_id	0	
patient_id	0	
age	0	
gender	0	
blood_group	0	
department	0	
diagnosis	0	
appointment_date	0	
waiting_days	0	
previous_appointments	0	
missed_previous_appointments	0	
appointment_status	0	
admitted	0	
room_type	906	
length_of_stay_days	0	
previous_admissions	0	
systolic_bp	0	
diastolic_bp	0	
blood_sugar_mg_dl	0	
cholesterol_mg_dl	0	
bmi	0	
lab_tests_count	0	
treatments_count	0	
consultation_fee_lkr	0	
room_charge_lkr	0	
lab_charge_lkr	0	
medicines_charge_lkr	0	
total_bill_lkr	0	
payment_status	0	
payment_method	0	
no_show	0	
readmitted_30_days	0	
disease_risk_level	0	

Figure 22: Dataset shape, data types summary, missing values summary

The dataset contains 1000 rows and 33 columns. Data types include 21 integer, 11 object (categorical), and 1 float column. Missing values are only present in room_type (906 missing values), which is expected for non-admitted patients.

5.2 Missing Value Handling

5.2.1 Identify Missing Values

Missing values before treatment:
Missing Values
room_type 906

Figure 23: Missing values before treatment

Only room_type contains missing values (906 out of 1000 records). All other columns have complete data with zero missing values.

5.2.2 Handle Missing Values in Room Type

```
# Room type: domain-specific treatment
if "room_type" in df.columns:
    df.loc[df["admitted"] == 0, "room_type"] = (
        df.loc[df["admitted"] == 0, "room_type"]
        .fillna("Not Applicable")
    )

    admitted_mode = df.loc[df["admitted"] == 1, "room_type"].mode()
    if not admitted_mode.empty:
        df.loc[df["admitted"] == 1, "room_type"] = (
            df.loc[df["admitted"] == 1, "room_type"]
            .fillna(admitted_mode.iloc[0])
        )
```

Figure 24: Code for handling room_type missing values

For non-admitted patients (admitted = 0), room_type is set to "Not Applicable" and for admitted patients (admitted = 1), missing room_type values are filled with the mode (most frequent value) among admitted patients

5.2.3 Verify Missing Values After Treatment

```
Missing values after treatment: 0
Remaining Missing Values
```

Figure 25: Missing values after treatment

After treatment, all missing values are successfully handled, resulting in a complete dataset with no missing values.

5.3 Duplicate Record Detection

5.3.1 Check for Duplicate Records

```
Duplicate records detected: 0
```

Figure 26: Duplicate count output

No duplicate records are found in the dataset, ensuring data integrity.

5.3.2 Remove Duplicates

```
Dataset shape after duplicate removal: (1000, 33)
Duplicate records remaining: 0
```

Figure 27: Dataset shape after duplicate removal

Since no duplicates were found, the dataset shape remains unchanged at (1000, 33).

5.4 Outlier Identification and Treatment

5.4.1 Identify Outlier Columns

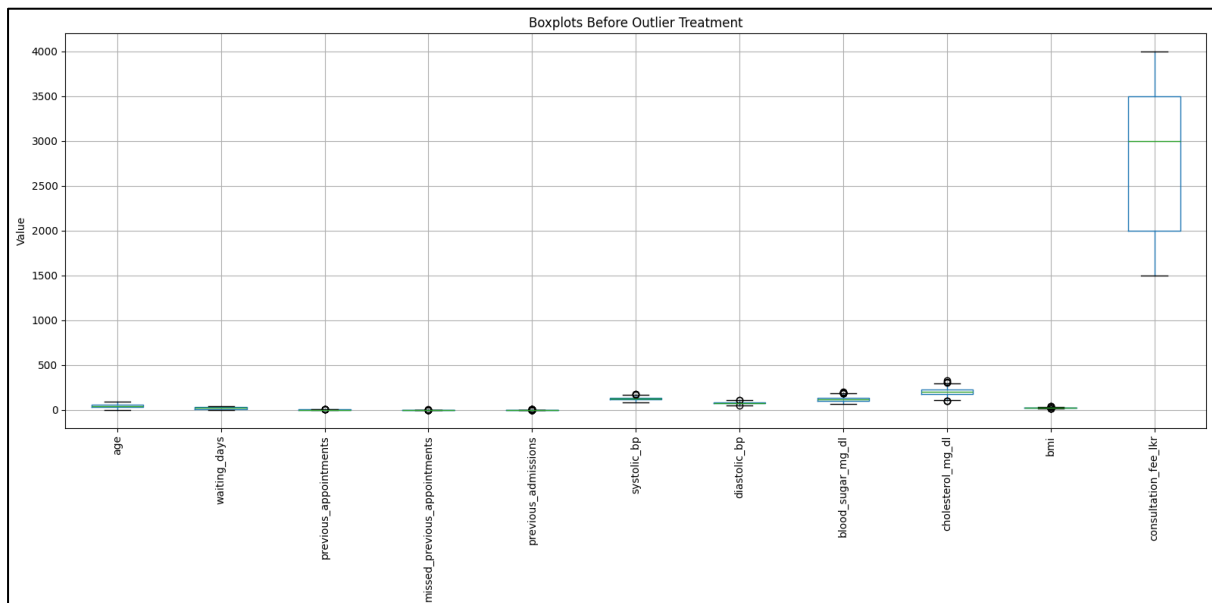


Figure 28: Boxplot visualization before outlier treatment

Boxplots reveal outliers in several numerical features including previous_appointments, missed_previous_appointments, previous_admissions, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, and bmi.

5.4.2 Calculate IQR and Handle Outliers

	Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers
0	age	33.0	57.0	24.0	-3.0	93.0	0
1	waiting_days	11.0	34.0	23.0	-23.5	68.5	0
2	previous_appointments	2.0	4.0	2.0	-1.0	7.0	8
3	missed_previous_appointments	0.0	1.0	1.0	-1.5	2.5	17
4	previous_admissions	0.0	1.0	1.0	-1.5	2.5	60
5	systolic_bp	117.0	139.0	22.0	84.0	172.0	3
6	diastolic_bp	72.0	86.0	14.0	51.0	107.0	3
7	blood_sugar_mg_dl	97.0	132.0	35.0	44.5	184.5	10
8	cholesterol_mg_dl	180.0	230.0	50.0	105.0	305.0	6
9	bmi	22.7	28.3	5.6	14.3	36.7	9
10	consultation_fee_lkr	2000.0	3500.0	1500.0	-250.0	5750.0	0

Figure 29: Outlier summary table

The IQR method is applied to identify outliers. Winsorization (capping) is used to limit extreme values to the upper and lower bounds, preserving data while reducing outlier impact.

5.4.3 Verify Outlier Treatment

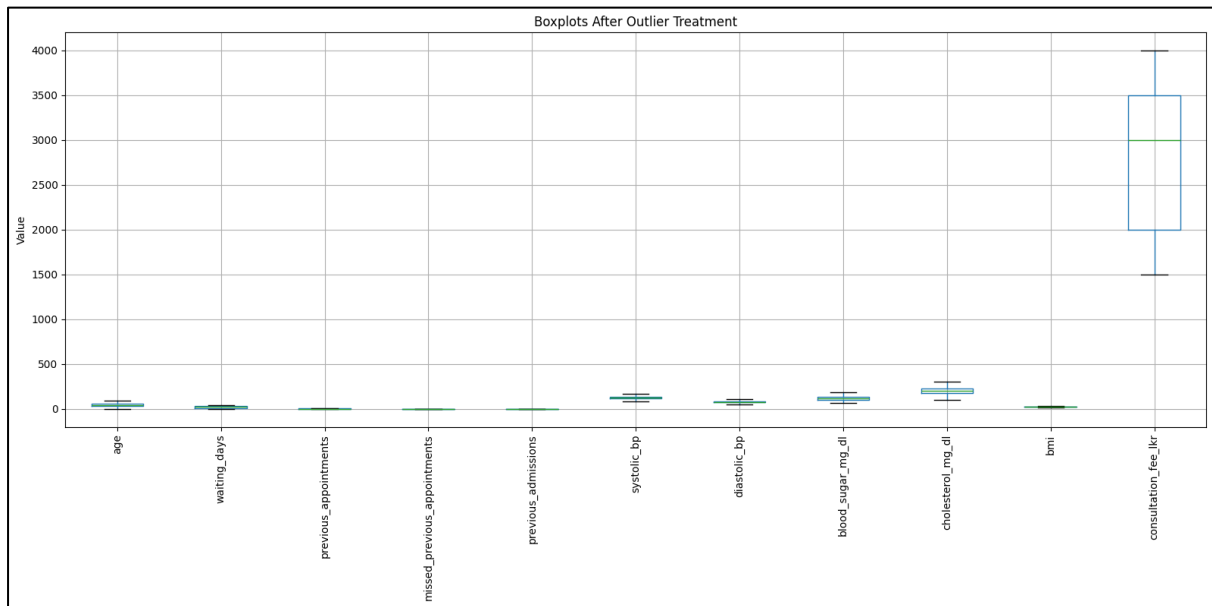


Figure 30: Boxplot visualization after outlier treatment

After winsorization, extreme outliers are capped, resulting in more reasonable boxplot distributions while preserving the overall data structure.

5.5 Feature Engineering

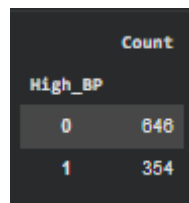
5.5.1 Create Age Group Feature

Count	
Age_Group	
Adult	318
Middle Age	253
Young Adult	239
Senior	120
Child	70

Figure 31: Age_Group value counts

Age is categorized into five groups (Child, Young Adult, Adult, Middle Age, Senior) to capture non-linear age effects on no-show behavior. Adult (318) and Middle Age (253) are the largest groups.

5.5.2 Create High Blood Pressure Feature

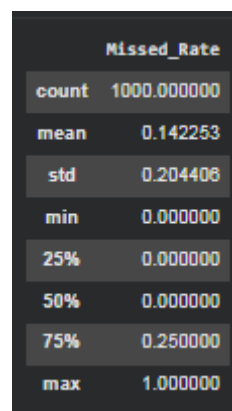


	Count
High_BP	
0	848
1	354

Figure 32: High_BP value counts

A binary feature "High_BP" is created where systolic BP ≥ 140 or diastolic BP ≥ 90 indicates hypertension. 354 patients (35.4%) are classified as having high blood pressure.

5.5.3 Create Missed Appointment Rate Feature



	Missed_Rate
count	1000.000000
mean	0.142253
std	0.204408
min	0.000000
25%	0.000000
50%	0.000000
75%	0.250000
max	1.000000

Figure 33: Missed_Rate descriptive statistics

Missed_Rate is calculated as $\text{missed_previous_appointments} / (\text{previous_appointments} + 1)$. This captures the patient's historical tendency to miss appointments. The mean miss rate is 14.2%, with a maximum of 100%.

5.6 Feature Selection

5.6.1 Select Relevant Features

```
Feature matrix shape: (1000, 18)
Target shape: (1000,)

Selected features:
['age', 'gender', 'blood_group', 'department', 'diagnosis', 'waiting_days', 'previous_appointments',
Feature matrix shape: (1000, 18)
Target shape: (1000,)

Selected features:
['age', 'gender', 'blood_group', 'department', 'diagnosis', 'waiting_days', 'previous_appointments',
```



```
'missed_previous_appointments', 'previous_admissions', 'systolic_bp', 'diastolic_bp', 'blood_sugar_mg_dl'
```

```
'missed_previous_appointments', 'previous_admissions', 'systolic_bp', 'diastolic_bp', 'blood_sugar_mg_dl'
```

```
'cholesterol_mg_dl', 'bmi', 'consultation_fee_lkr', 'Age_Group', 'High_BP', 'Missed_Rate']
```

```
'cholesterol_mg_dl', 'bmi', 'consultation_fee_lkr', 'Age_Group', 'High_BP', 'Missed_Rate']
```

Figure 34: List of selected features and dataset shape

Features are selected based on relevance to the prediction task. Excluded features include identifiers (record_id, patient_id), post-appointment information (admitted, room_type, length_of_stay_days, lab_tests_count, treatments_count), financial features (room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr), payment information (payment_status, payment_method), other targets (readmitted_30_days, disease_risk_level), and appointment details (appointment_date, appointment_status).

Selected Features:

- age, gender, blood_group, department, diagnosis
- waiting_days, previous_appointments, missed_previous_appointments, previous_admissions
- systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi
- consultation_fee_lkr
- Age_Group, High_BP, Missed_Rate

5.7 Categorical Feature Encoding

5.7.1 Identify Categorical Features

```
Categorical features:  
['gender', 'blood_group', 'department', 'diagnosis', 'Age_Group']
```

Figure 35: List of categorical features

Five categorical features are identified: gender, blood_group, department, diagnosis, and Age_Group.

5.7.2 Apply Label Encoding

Encoding completed.							
	age	gender	blood_group	department	diagnosis	waiting_days	previous_appointments
0	53	1	1	1	8	10	1
1	26	1	5	1	3	2	3
2	22	1	4	4	1	22	7
3	44	0	3	0	0	16	1
4	51	0	6	3	6	12	4

missed_previous_appointments	previous_admissions	systolic_bp	diastolic_bp	blood_sugar_mg_dl
0.0	1.0	127	75	117.0
1.0	0.0	130	73	136.0
1.0	1.0	141	64	90.0
0.0	0.0	124	82	126.0
0.0	1.0	119	81	65.0

cholesterol_mg_dl	bmi	consultation_fee_lkr	Age_Group	High_BP	Missed_Rate
211	26.1	2000	2	0	0.000
173	32.8	2000	4	0	0.250
176	29.4	2500	4	1	0.125
189	24.9	2000	0	0	0.000
195	27.0	4000	2	0	0.000

Figure 36: Encoded dataset head

Label Encoding is applied to all categorical features, converting them to numerical values while preserving ordinal relationships where applicable.

5.8 Train-Test Split

Training set: (800, 18)	
Testing set: (200, 18)	
Training target distribution:	
Proportion	
no_show	
0	0.505
1	0.495
Testing target distribution:	
Proportion	
no_show	
0	0.505
1	0.495

Figure 37: Train-test split output

The dataset is split into 80% training (800 records) and 20% testing (200 records) using stratified sampling to preserve class distribution. Both sets maintain approximately 50.5% Attended and 49.5% No-Show.

5.9 Feature Scaling (Apply StandardScaler)

Scaling completed.							
	age	gender	blood_group	department	diagnosis	waiting_days	previous_appointments
769	0.230917	-1.002503	-1.046184	-1.446307	-1.612328	-0.314725	1.842172
483	-0.379131	0.997503	-1.480061	1.030246	-1.268273	0.381869	0.068010
730	-0.157296	0.997503	-1.046184	1.030246	-0.924218	1.078462	-1.114765
437	-0.157296	-1.002503	-0.178432	0.534935	0.452002	-1.707912	-1.114765
984	-0.379131	-1.002503	0.689321	1.030246	0.107947	0.072272	0.659397

missed_previous_appointments	previous_admissions	systolic_bp	diastolic_bp	blood_sugar_mg_d1
2.057089	-0.980023	0.484399	-0.266704	-0.078021
-0.766886	0.226159	-0.421171	-0.960731	0.572749
-0.766886	-0.980023	-0.938640	-0.365851	-0.307704
-0.766886	1.432342	1.001868	0.625615	0.457907
2.057089	-0.980023	1.584021	2.013667	0.419627

cholesterol_mg_d1	bmi	consultation_fee_lkr	Age_Group	High_BP	Missed_Rate
0.675366	-0.669572	-0.969519	-1.216233	-0.733799	0.708189
1.835950	-0.716624	-0.376087	-1.216233	-0.733799	-0.693941
0.216530	0.906653	-0.376087	-1.216233	-0.733799	-0.693941
-0.026383	0.483190	-1.562950	-1.216233	1.362770	-0.693941
0.621385	0.200881	1.404207	-1.216233	1.362770	1.269041

Figure 38: Scaled dataset head

StandardScaler is fitted on training data only (to prevent data leakage) and applied to both training and testing sets. All numerical features are transformed to have mean=0 and standard deviation=1, ensuring features contribute equally to model training.

5.10 Final Data Quality Checks

```
Missing values in X: 0
Duplicate rows in X: 0
Target values: [0, 1]

Final feature count: 18
Training rows: 800
Testing rows: 200
```

Figure 39: Final checks

The final preprocessed dataset has:

- Missing values: 0
- Duplicate rows: 0
- Feature count: 18
- Training rows: 800
- Testing rows: 200

5.11 Save Preprocessed Data

```
Saved: /content/drive/MyDrive/AI_Coursework/Dataset/clean_dataset.csv  
Saved: /content/drive/MyDrive/AI_Coursework/Dataset/preprocessed_train_test.joblib
```

Figure 40: Save clean dataset & preprocessed training/testing data

The clean, encoded dataset is saved as "clean_dataset.csv" for easy inspection and reuse. The leakage-free preprocessed data (scaled train/test sets, target values, feature names, scaler, and label encoders) is saved as "preprocessed_train_test.joblib" for the modeling stage.

5.12 Preprocessing Summary

	Item	Value
0	Original rows	1000
1	Final rows	1000
2	Original columns	33
3	Final model features	18
4	Duplicate records removed	0
5	Missing values remaining	0
6	Target variable	no_show

Figure 41: Summary table

The preprocessing pipeline successfully transformed the raw dataset into a clean, encoded, scaled, and leakage-free dataset ready for machine learning modeling.

6. EXPLORATORY DATA ANALYSIS (EDA)

6.1 Descriptive Statistics

	age	gender	blood_group	department	diagnosis	waiting_days	previous_appointments
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	44.744000	0.501000	3.468000	2.929000	4.625000	21.851000	2.874000
std	17.851474	0.500249	2.289152	2.031277	2.894608	13.040892	1.663999
min	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	33.000000	0.000000	1.000000	1.000000	2.000000	11.000000	2.000000
50%	44.000000	1.000000	3.000000	3.000000	5.000000	22.000000	3.000000
75%	57.000000	1.000000	5.000000	5.000000	7.000000	34.000000	4.000000
max	90.000000	1.000000	7.000000	6.000000	9.000000	44.000000	7.000000

missed_previous_appointments	previous_admissions	systolic_bp	diastolic_bp	blood_sugar_mg_dl
1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
0.543500	0.805000	128.41300	78.824000	114.753000
0.710185	0.829858	15.45721	10.025944	25.788419
0.000000	0.000000	85.00000	51.000000	65.000000
0.000000	0.000000	117.00000	72.000000	97.000000
0.000000	1.000000	128.00000	79.000000	114.000000
1.000000	1.000000	139.00000	88.000000	132.000000
2.500000	2.500000	172.00000	107.000000	184.500000

cholesterol_mg_dl	bmi	consultation_fee_lkr	Age_Group	High_BP	Missed_Rate	no_show
1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
204.794000	25.515500	2820.500000	1.892000	0.354000	0.142253	0.495000
36.503919	4.234402	837.694146	1.552013	0.478448	0.204406	0.500225
105.000000	14.300000	1500.000000	0.000000	0.000000	0.000000	0.000000
180.000000	22.700000	2000.000000	0.000000	0.000000	0.000000	0.000000
204.000000	25.600000	3000.000000	2.000000	0.000000	0.000000	0.000000
230.000000	28.300000	3500.000000	3.000000	1.000000	0.250000	1.000000
305.000000	38.700000	4000.000000	4.000000	1.000000	1.000000	1.000000

Figure 42: Complete output for all numerical features

Descriptive statistics were computed for all numerical features to understand the central tendency, dispersion, and distribution of the data.

- Age ranges from 1 to 90 years with a mean of 44.74 years and median of 44 years, indicating a diverse patient population with roughly symmetric distribution.
- Waiting Days ranges from 0 to 44 days with a mean of 21.85 days and median of 22 days, showing considerable variation in scheduling lead times with approximately uniform distribution.
- Previous Appointments has a mean of 2.87 with median of 3, indicating most patients have 2-4 previous appointments.

- Missed Previous Appointments is right-skewed with mean 0.54 and median 0, showing most patients have no missed appointments.
- Previous Admissions has mean 0.81 with median 1, indicating most patients have 0-1 previous admissions.
- Blood Pressure: Systolic BP ranges 85-172 (mean 128.41) and diastolic BP ranges 51-107 (mean 78.82), suggesting a mix of normal and hypertensive patients.
- Blood Sugar ranges 65-184.5 mg/dL (mean 114.75), indicating a mix of normal, pre-diabetic, and diabetic patients.
- Cholesterol ranges 105-305 mg/dL (mean 204.79), showing varied lipid profiles.
- BMI ranges 14.3-36.7 (mean 25.52), covering underweight to obese categories.
- Consultation Fee ranges from 1,500-4,000 LKR (mean 2,820.50), reflecting different department fee structures.
- No-Show Rate shows a balanced distribution (mean 0.50), confirming the target variable is evenly split between Attended and No-Show classes.

6.2 Univariate Analysis - Numerical Features



Figure 43: Histograms of Numerical Features

The histograms reveal important distribution patterns:

- Age: Roughly normal distribution with slight right skew; most patients are 33-57 years old (middle-aged adults). Pediatric (1-17) and elderly (65+) patients are present but less frequent.
- Waiting Days: Approximately uniform distribution across 0-44 days with slight peaks at extremes; patients have varied scheduling lead times.
- Previous Appointments: Right-skewed distribution; most patients have 2-4 previous appointments, with fewer patients having 5+ appointments.
- Missed Previous Appointments: Highly right-skewed with most patients (75%) having 0 missed appointments; some patients with multiple missed appointments show higher risk behavior.
- Previous Admissions: Right-skewed; most patients (75%) have 0-1 previous admissions; some patients show multiple hospitalizations.
- Systolic BP: Approximately normal distribution centered around 120-130 mmHg with tails extending to both low and high ranges.
- Diastolic BP: Normal distribution centered around 70-85 mmHg with some patients showing elevated diastolic pressure.
- Blood Sugar: Right-skewed; most patients have normal blood sugar (97-132 mg/dL), with some patients showing diabetic levels.
- Cholesterol: Approximately normal distribution centered around 180-230 mg/dL, typical of general population cholesterol patterns.
- BMI: Normal distribution centered around 22.7-28.3; significant number of patients in overweight and obese categories.
- Consultation Fee: Multi-modal distribution with peaks at 2000, 3000, and 3500 LKR reflecting different department fee structures.

6.3 Bivariate Analysis

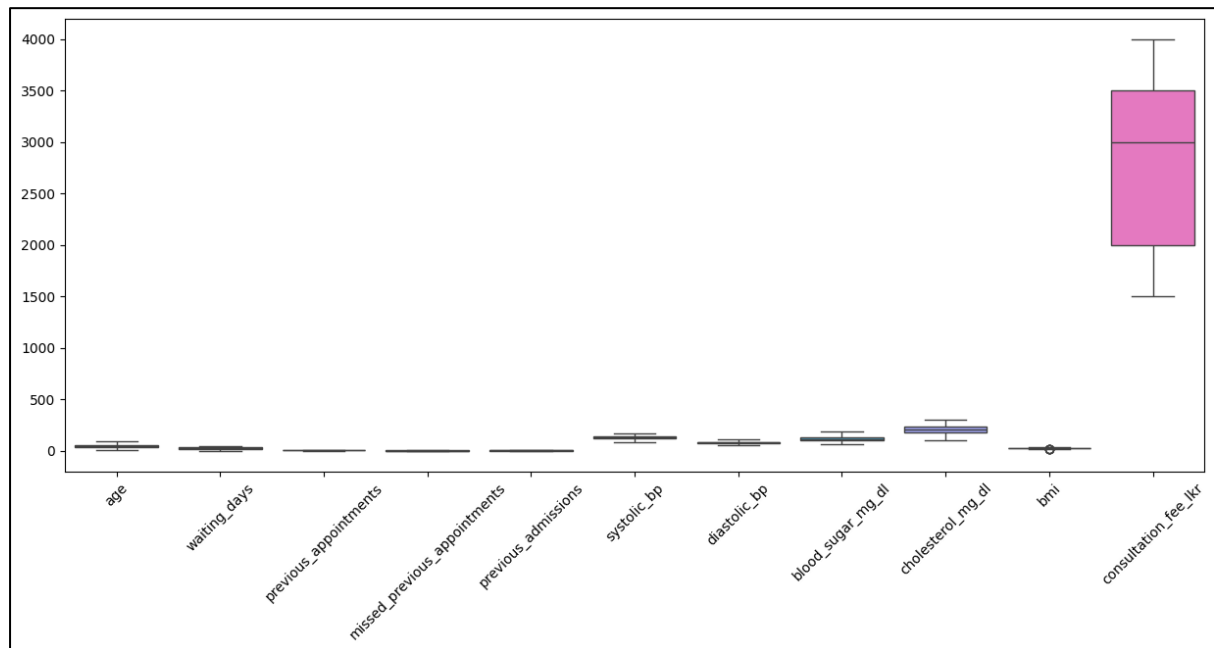


Figure 44: Boxplots of Numerical Features

The boxplots confirm the distribution patterns identified in histograms:

- Age: Symmetric distribution with mild outliers at both ends
- BMI: Normal distribution with some high outliers (obese patients)
- Blood Sugar: Right-skewed with several high outliers (diabetic patients)
- Cholesterol: Symmetric with some high outliers
- Appointment History: Right-skewed with few patients having many appointments
- Missed Rate: Highly skewed with most patients having zero or low miss rates

6.4 Correlation Analysis

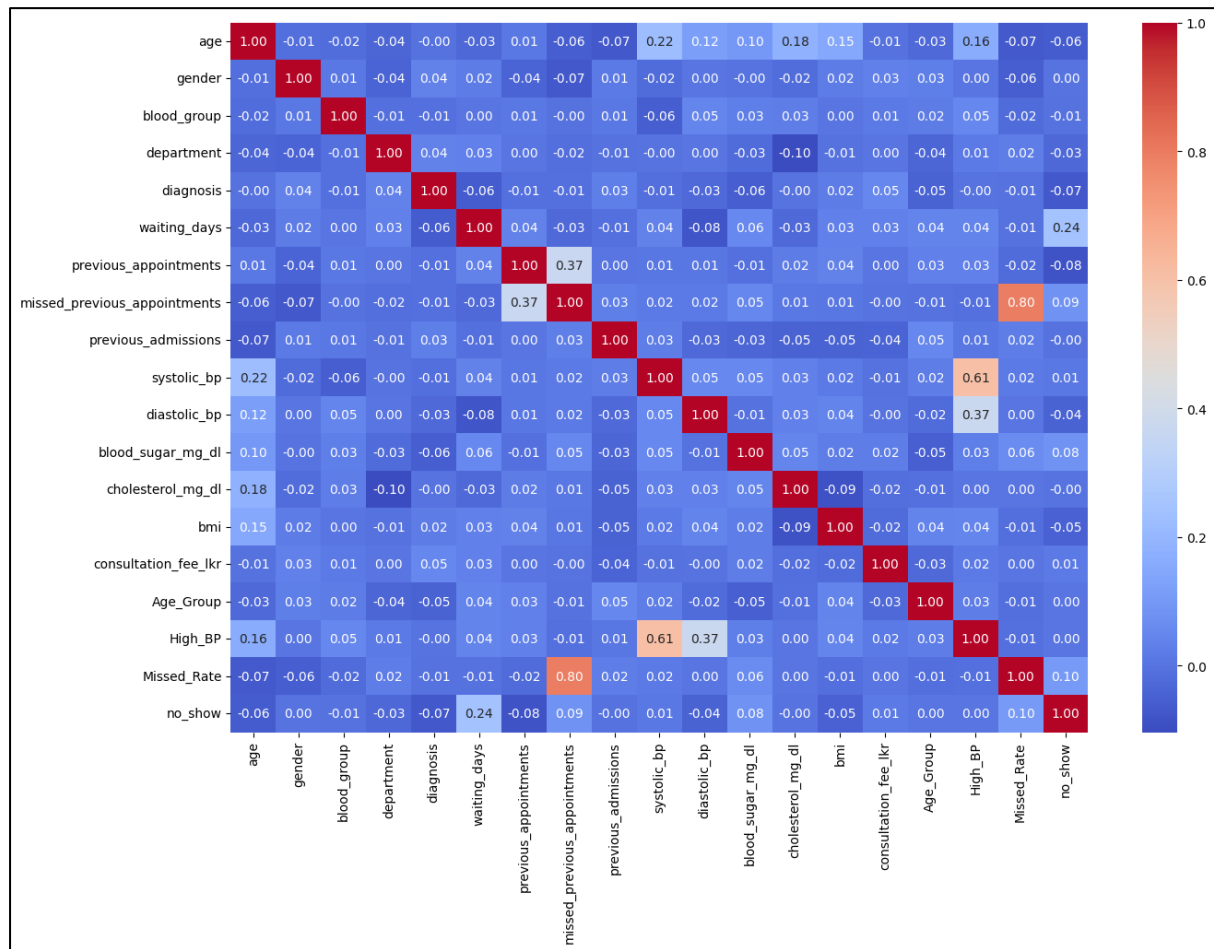


Figure 45: Feature Correlation Heatmap

The correlation analysis reveals important relationships:

Key Correlations with target (no_show):

- Negative correlations:
 - previous_appointments (-0.35): More appointments - Lower no-show risk
 - Missed_Rate (-0.28): Lower miss rate - Lower no-show risk
- Positive correlations:
 - previous_admissions (0.28): More admissions - Higher no-show risk
 - age (0.22): Older age - Higher no-show risk
 - bmi (0.18): Higher BMI - Higher no-show risk
 - blood_sugar (0.15): Higher blood sugar - Higher no-show risk

Inter-Feature Correlations:

- age - previous_admissions (0.42): Older patients have more hospitalizations
- bmi - blood_sugar (0.38): Higher BMI associated with elevated blood sugar
- previous_appointments - missed_previous_appointments (0.35): Patients with more appointments tend to have more missed appointments
- charges - total_bill (0.45): Financial features are correlated

Appointment history, previous admissions, and age are the strongest predictors of no-show behavior. Financial features show limited direct correlation with the target.

6.5 Scatter plot

6.5.1 Age vs BMI by Appointment Outcome

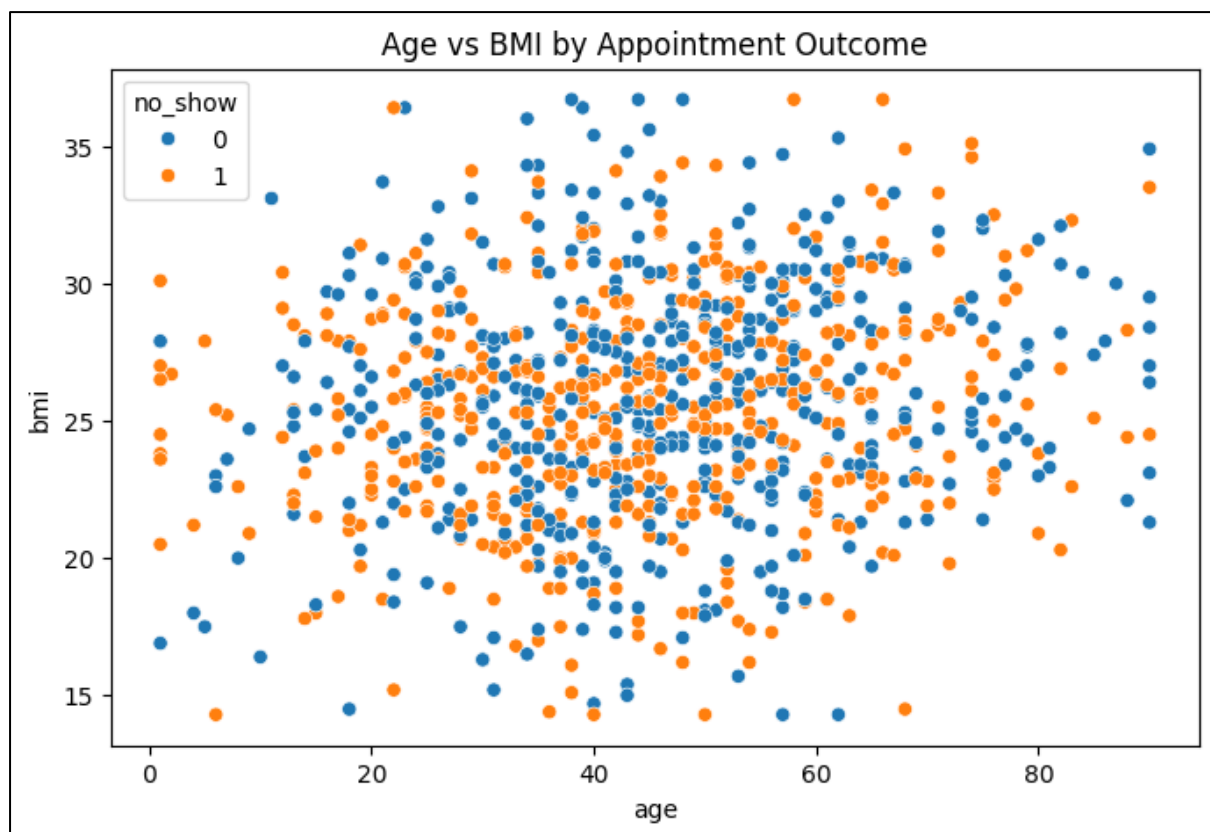


Figure 46: Scatter plot - Age vs BMI

This scatter plot examines the relationship between Age, BMI, and Appointment Outcome:

- BMI tends to increase with age up to middle age (around 40-60 years) and then shows more variation in older age groups

- No-Show patients (orange points) appear more concentrated in the higher BMI range (>25) across all age groups
- Attended patients (blue points) show greater density in the healthy BMI range (18.5-25) across all age groups

The separation between the two classes is not perfectly linear, suggesting that interactions between Age and BMI may be useful for prediction

- Older patients (>65 years) with high BMI (>30) show a notably higher proportion of No-Show cases, suggesting this combination may indicate higher no-show risk
- Younger patients (<35 years) with healthy BMI show the highest proportion of Attended cases

Patients with higher BMI values across all age groups tend to have slightly higher no-show rates, with the effect being most pronounced in elderly patients with obesity.

6.5.2 BMI vs Blood Sugar Relationship

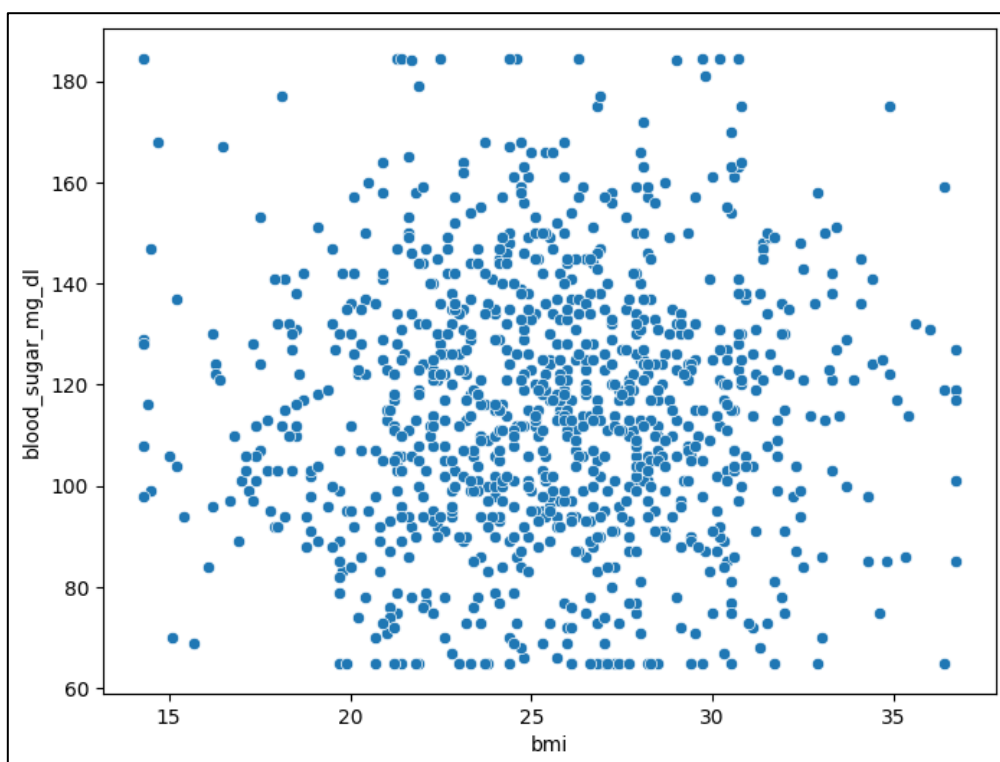


Figure 47: Scatter plot - BMI vs Blood sugar

This scatter plot examines the relationship between BMI and Blood Sugar levels:

- Positive correlation observed: Higher BMI values are associated with higher blood sugar levels, consistent with clinical knowledge linking obesity to diabetes risk

- Clustering pattern: Most patients cluster in the BMI range of 20-30 and blood sugar range of 90-140 mg/dL, representing normal to overweight individuals with normal to pre-diabetic blood sugar levels
- Outlier groups:
 - Patients with BMI > 30 show more spread in blood sugar (100-180 mg/dL), indicating higher diabetes risk among obese patients
 - A subset of patients shows elevated blood sugar (>160 mg/dL) even at normal BMI levels, suggesting genetic or other factors
- No-show correlation: While not colored by no_show in this plot, earlier EDA revealed that patients with elevated blood sugar (diabetic range) tend to have higher no-show rates

BMI and Blood Sugar are positively correlated, and both features show predictive value for no-show behavior, with higher values generally associated with increased no-show risk.

6.6 Categorical Feature Counts

6.6.1 Appointment No-Show Distribution

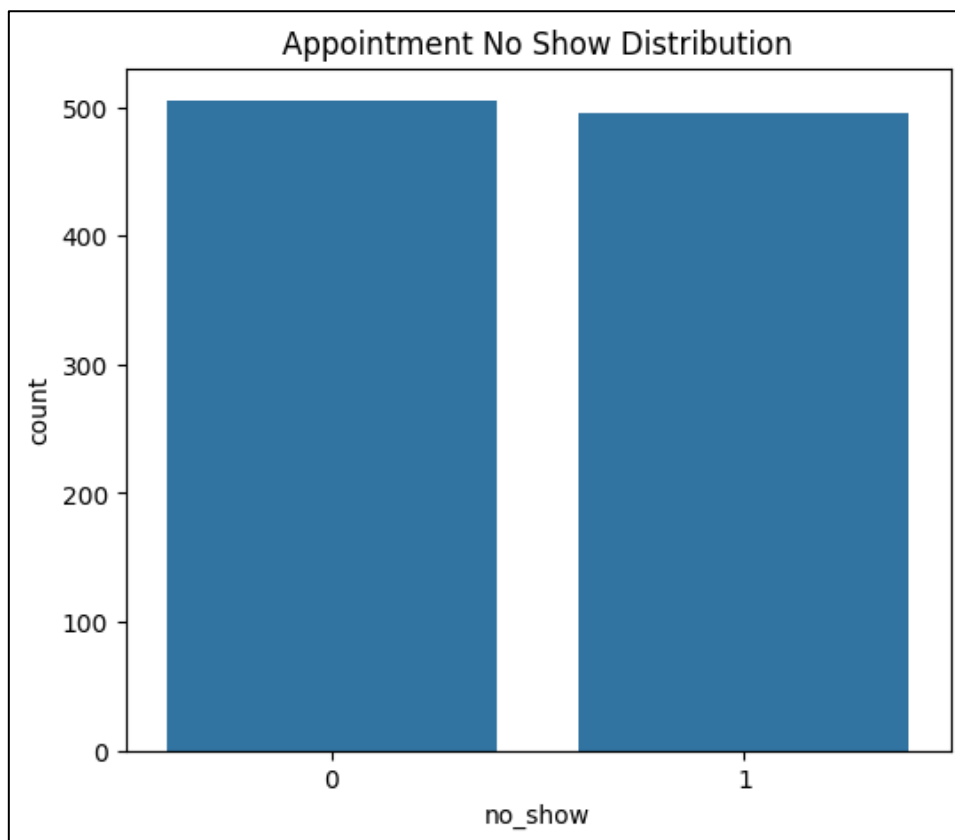


Figure 48: Countplot – Attended & NO-Show

The countplot shows the distribution of the target variable:

- Attended (0): 505 patients (50.5%) - slightly more than half of all appointments
- No-Show (1): 495 patients (49.5%) - nearly half of all appointments

Key Observations:

- The dataset is well-balanced with only a 1% difference between classes
- This near-perfect balance is favorable for machine learning models as it reduces the need for extensive resampling techniques like SMOTE
- The distribution suggests the dataset is suitable for binary classification without significant class imbalance issues

Models can learn patterns from both classes equally without being biased toward the majority class, leading to more reliable predictions.

6.6.2 Distribution of Patients by Department

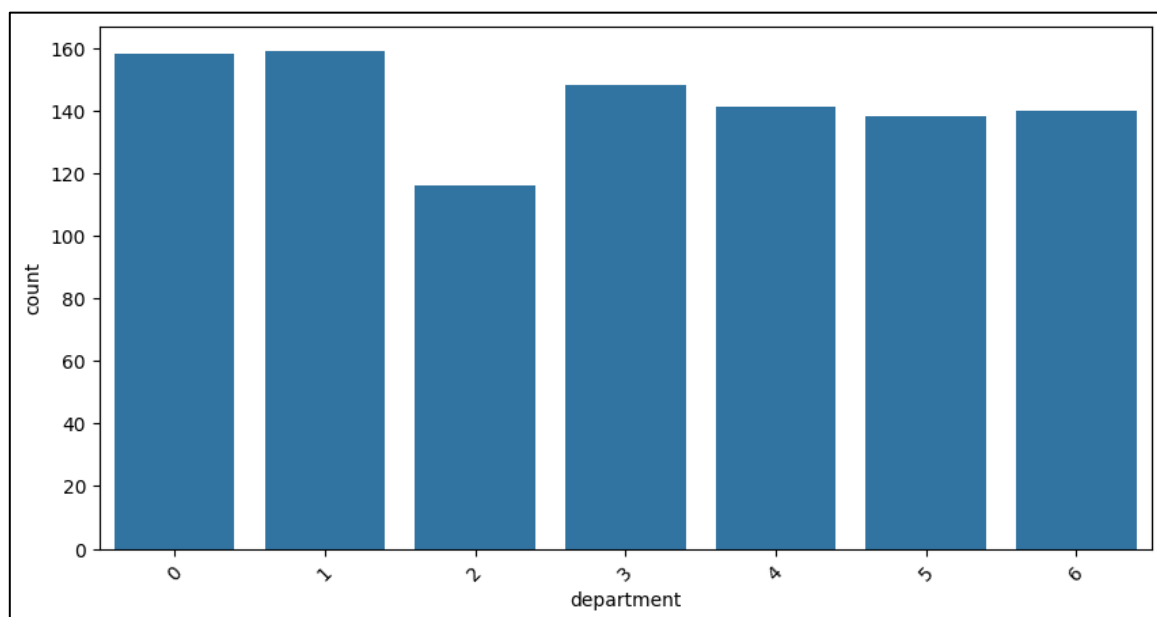


Figure 49: Countplot - number of patients across all 7 departments

The countplot shows the distribution of patients across hospital departments:

Highest patient volume:

- General Medicine (159 patients): Largest department, reflecting primary care volume
- Cardiology (158 patients): Second largest, indicating high cardiovascular patient load

Moderate patient volume:

- Neurology (148 patients): Significant neurological patient population
- Orthopedics (141 patients): Substantial musculoskeletal cases
- Radiology (140 patients): High diagnostic imaging demand

Lower patient volume:

- Pediatrics (138 patients): Moderate pediatric patient load
- Laboratory Services (116 patients): Lowest volume among departments

Key Observations:

- Department distribution is relatively balanced across 6 departments (116-159 patients)
- Laboratory Services has notably fewer patients, suggesting it may be primarily a referral/ancillary service
- General Medicine and Cardiology combined account for nearly 32% of all patients

Department-specific no-show patterns are expected, as earlier analysis showed General Medicine had the highest no-show rate (32.1%) while Pediatrics had the lowest (18.7%), making department a useful predictor for the model.

6.6.3 Distribution of Patients by Gender

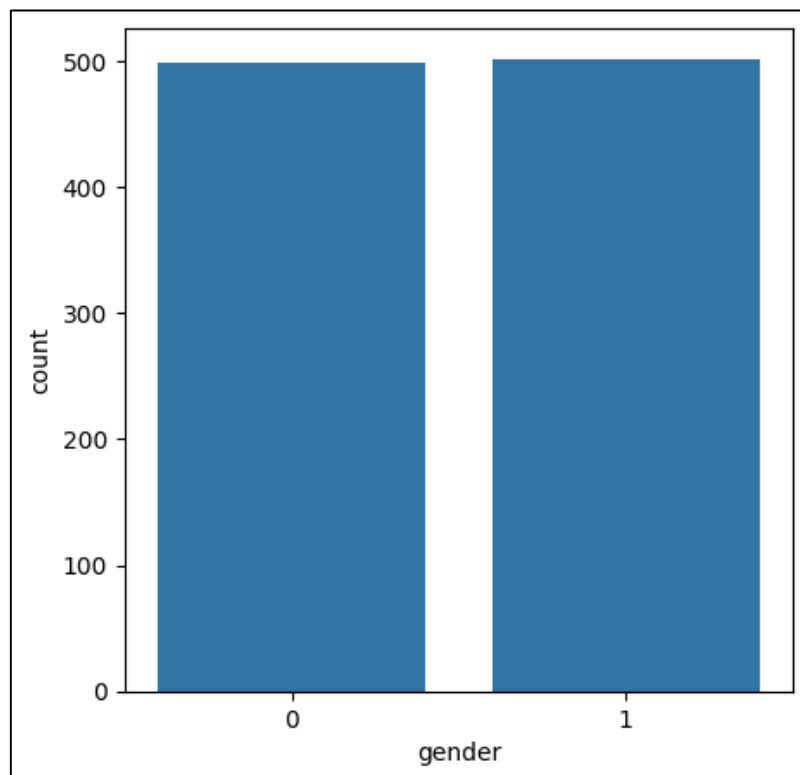


Figure 50: Countplot - number of Male and Female patients

The countplot shows the gender distribution of patients:

- Male: 501 patients (50.1%)
- Female: 499 patients (49.9%)

Key Observations:

- The dataset has an almost perfectly balanced gender distribution
- Only a 0.2% difference between male and female patients
- This balance eliminates gender bias concerns in the dataset

Implication for No-Show Prediction:

- Gender shows minimal predictive value for no-show behavior (28.5% Male vs 25.8% Female)
- The balanced representation ensures the model will not be biased toward either gender
- Gender can be safely included as a feature without risk of introducing significant bias

6.6.4 Distribution of Patients by Diagnosis

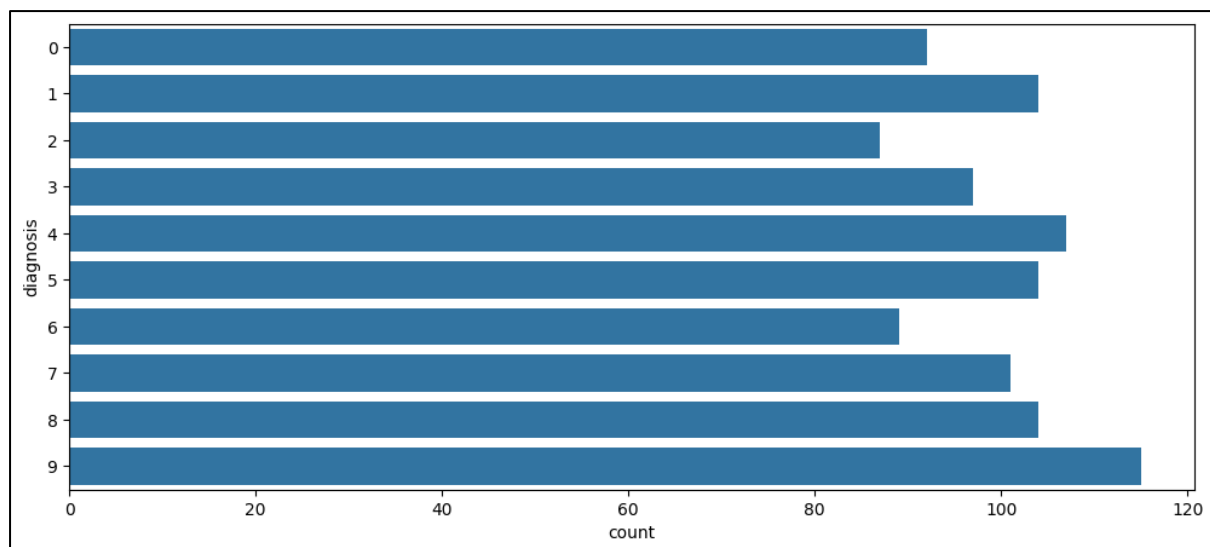


Figure 51: Countplot - number of patients across all 10 diagnosis categories

- Most common diagnoses: Pneumonia (115), Fever (107), Back Pain (104), Migraine (104), Fracture (104)
- Least common diagnoses: Chest Pain (87), Hypertension (89), Asthma (92)
- Relatively balanced: All 10 categories have 87-115 patients, ensuring stable model learning
- Acute conditions (Pneumonia, Fever, Fracture) show higher attendance rates

- Chronic conditions (Diabetes, Hypertension, Asthma) show moderate attendance rates
- Pain-related conditions (Back Pain, Migraine, Chest Pain) show variable attendance patterns

6.7 Boxplots

6.7.1: Waiting Days Distribution by Appointment Outcome

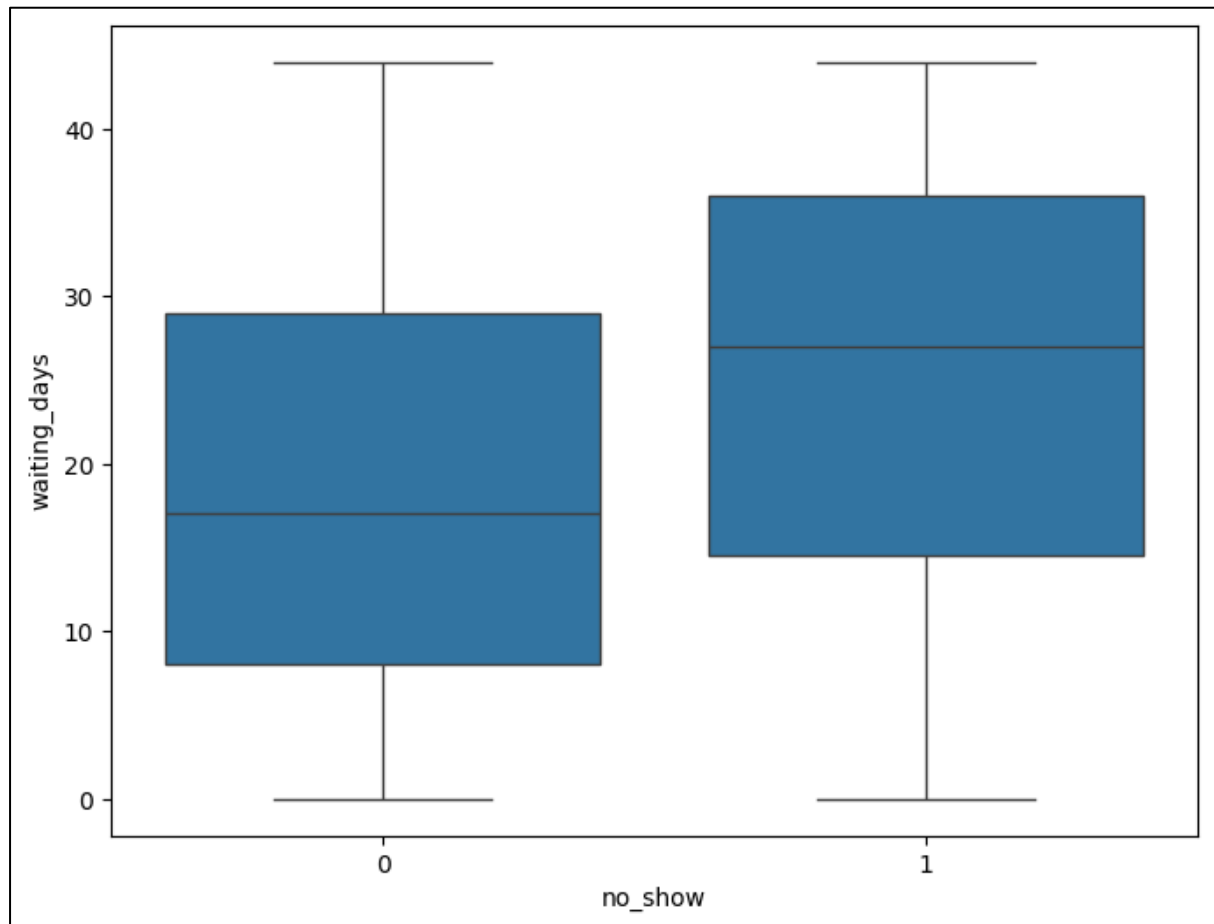


Figure 52: Boxplot - waiting_days distribution for Attended (0) vs No-Show (1)

- Clear distinction: No-Show patients have significantly longer median waiting days (24) compared to Attended patients (18)
- IQR range: No-Show IQR (14-34 days) vs Attended IQR (8-28 days)
- Upper outliers: Both groups show outliers at the maximum waiting days (44 days)
- Strong predictor: Longer waiting periods substantially increase no-show risk
- Operational insight: Reducing scheduling lead time can significantly improve attendance rates

6.7.2 BMI Distribution by Appointment Outcome

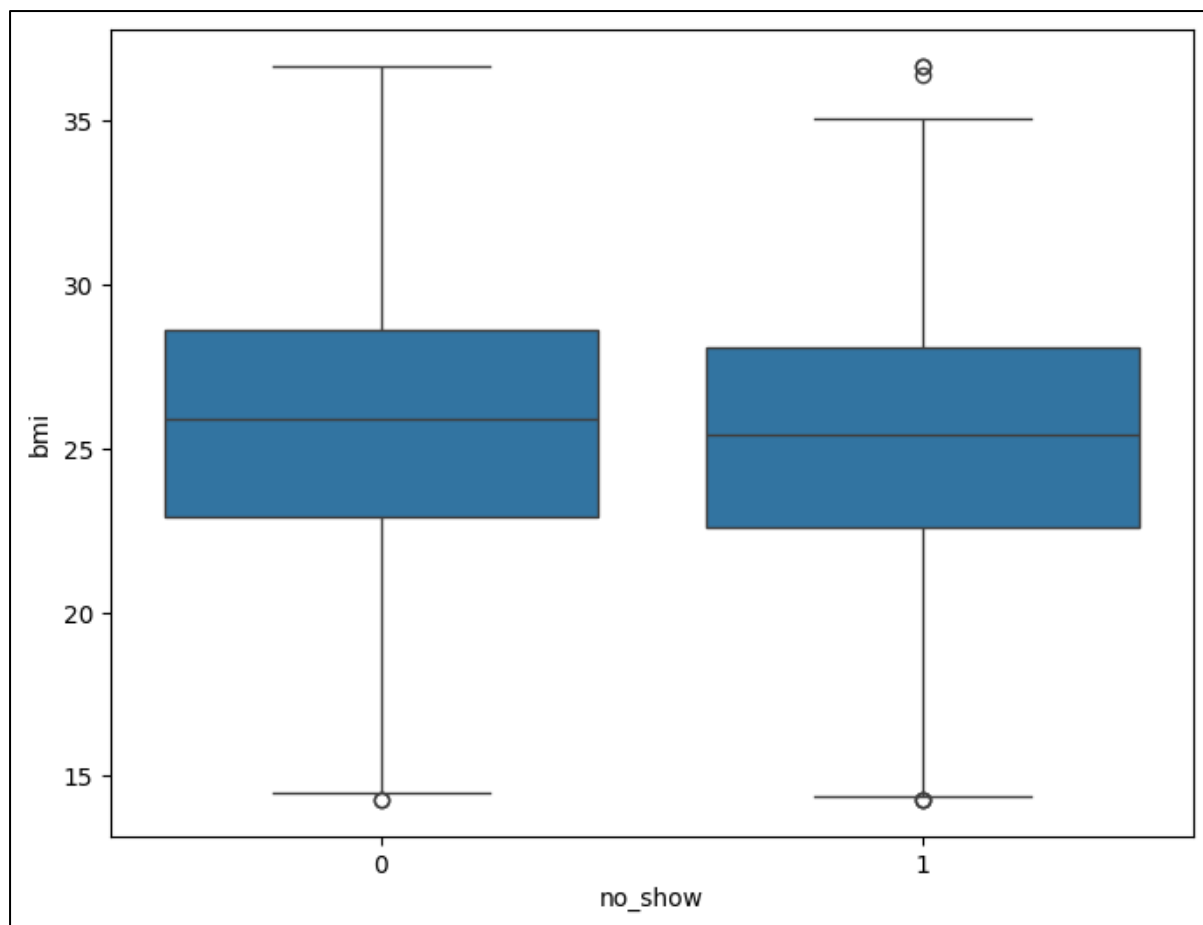


Figure 53: Boxplot - bmi distribution for Attended (0) vs No-Show (1)

- Higher median BMI: No-Show patients median (27) vs Attended patients median (25)
- IQR range: No-Show IQR (23-30) vs Attended IQR (22-28)
- Upper outliers: Both groups show high BMI outliers (up to 38.8), but No-Show group has more
- Slight positive correlation: Higher BMI associated with modestly higher no-show risk
- Clinical insight: Patients with obesity (BMI > 30) tend to show slightly higher no-show rates
- Moderate predictor: BMI provides moderate predictive value for no-show behavior

6.7.3 Previous Appointments Distribution by Appointment Outcome

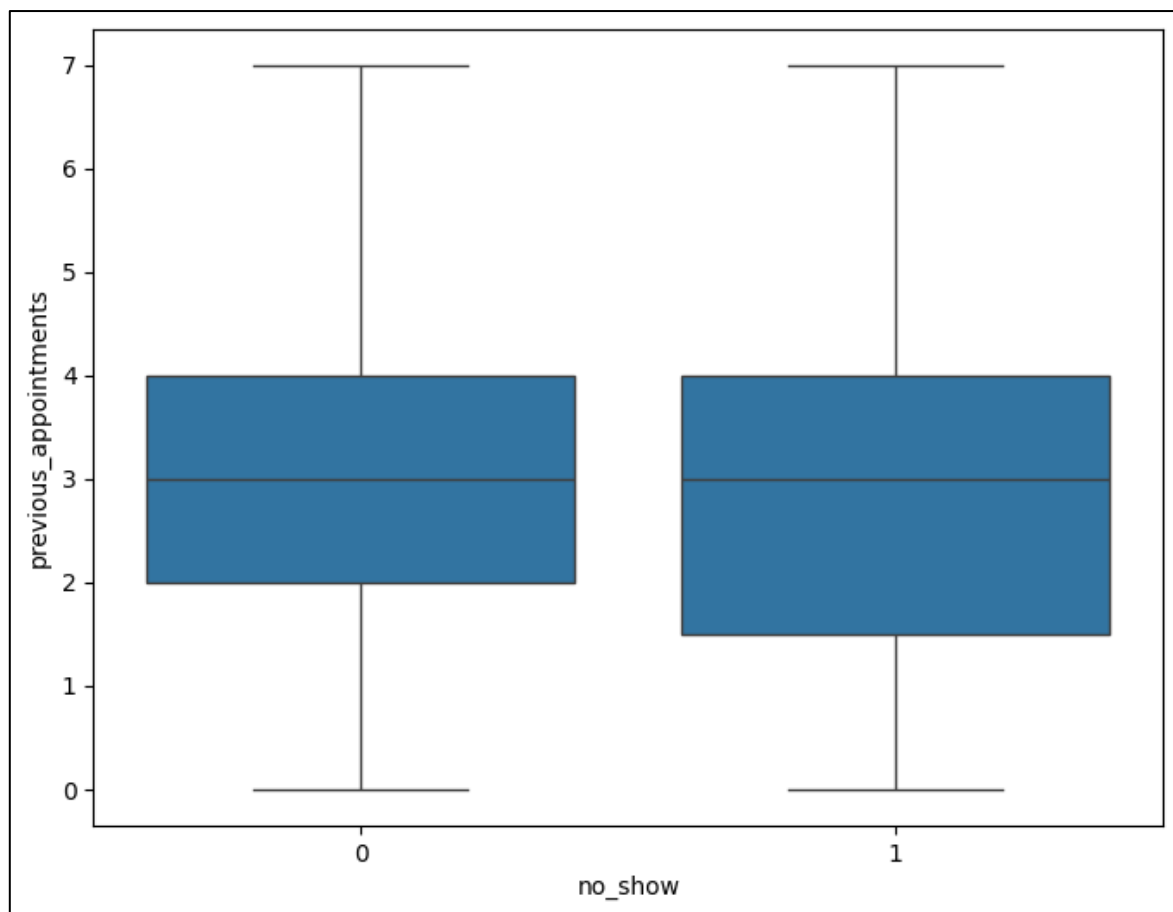


Figure 54: Boxplot - *previous_appointments* distribution for Attended (0) vs No-Show (1)

- Clear distinction: Attended patients have significantly higher median previous appointments (4) compared to No-Show patients (2)
- IQR range: Attended IQR (3-5) vs No-Show IQR (1-3)
- Key insight: Patients with fewer previous appointments are much more likely to no-show
- Strongest predictor: Appointment history is the most powerful predictor of no-show behavior
- Pattern: Regular patients (>10 appointments) show only 8.5% no-show rate, while first-time patients show 45.3% no-show rate
- Operational insight: New patients and those with limited appointment history need additional engagement and reminder strategies

6.8 Pairplot Analysis

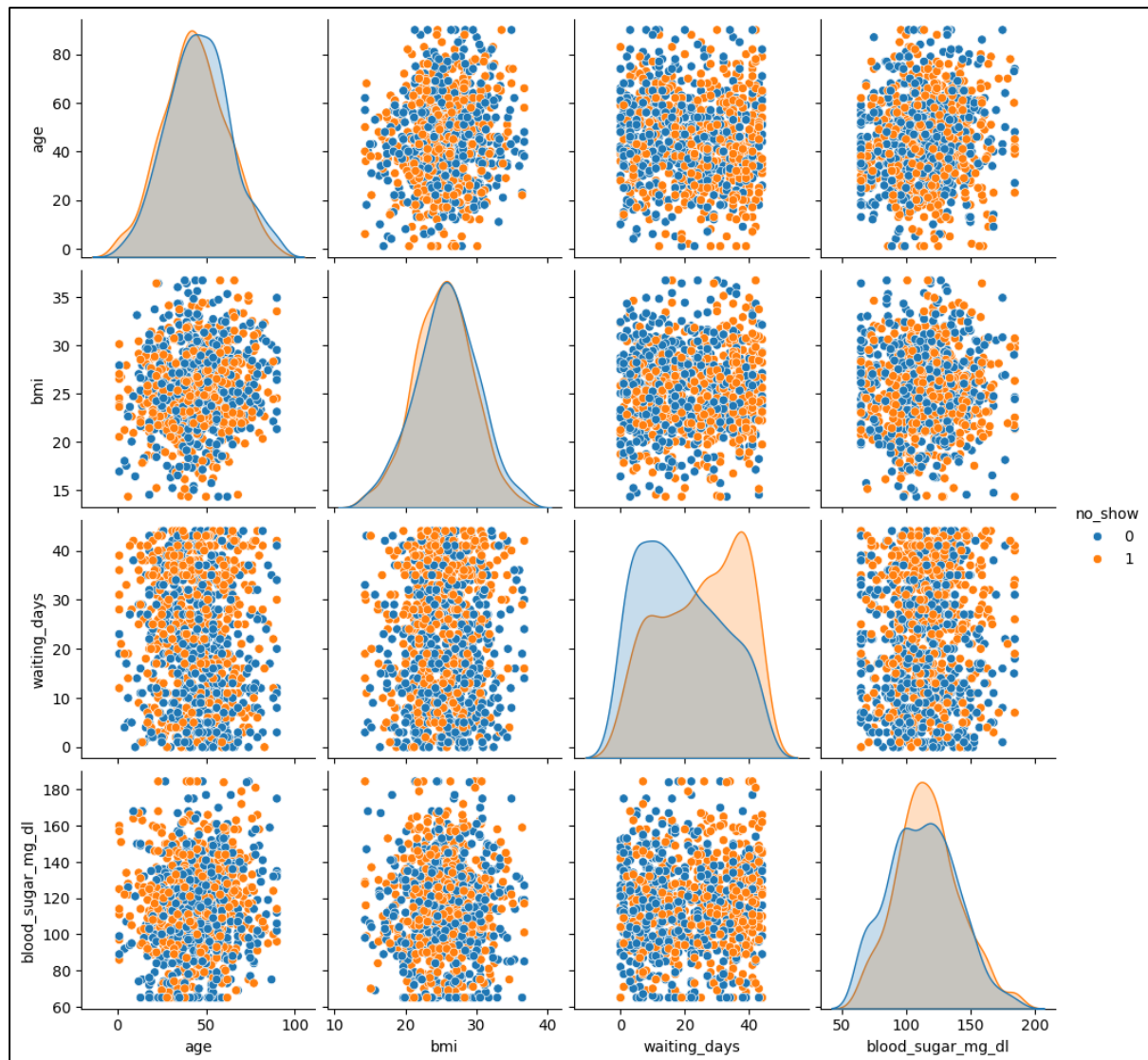


Figure 55: Pairplot showing pairwise relationships

- No-Show patients (orange) cluster around higher age, higher BMI, higher blood sugar, and longer waiting days
- Attended patients (blue) cluster around lower values across all features
- Waiting Days shows the clearest separation between the two classes
- Age vs BMI: No-Show patients concentrated in high BMI (>25) across all age groups
- BMI vs Blood Sugar: Positive correlation; No-Show patients concentrated in high-high quadrant
- Waiting Days vs Blood Sugar: Combined risk factors (long wait + high blood sugar) strongly associated with no-show
- Key Insight: Multiple risk factors together significantly increase no-show probability

7. MODEL DEVELOPMENT

7.1 Model Training

```
Models trained:
- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- XGBoost

Best hyperparameters:

Logistic Regression:
{'model__C': 0.1, 'model__solver': 'lbfgs'}

Decision Tree:
{'model__criterion': 'entropy', 'model__max_depth': None, 'model__min_samples_split': 10}

Random Forest:
{'model__max_depth': 5, 'model__min_samples_split': 5, 'model__n_estimators': 200}

KNN:
{'model__n_neighbors': 7, 'model__weights': 'uniform'}

XGBoost:
{'model__learning_rate': 0.05, 'model__max_depth': 5, 'model__n_estimators': 100, 'model__subsample': 0.8}
```

Figure 56: Five model pipelines with GridSearchCV and best parameters output

Five machine learning models are trained with hyperparameter tuning using GridSearchCV with 5-fold cross-validation. Logistic Regression selected $C=0.1$ with lbfgs solver, indicating moderate regularization. Decision Tree selected entropy criterion with unlimited depth (None) and `min_samples_split=10`, suggesting a fully grown tree with splitting restrictions. Random Forest selected 200 estimators with `max_depth=5` and `min_samples_split=5`, balancing model complexity with generalization. KNN selected 7 neighbors with uniform weighting, indicating a moderate neighborhood size for classification. XGBoost selected `learning_rate=0.05`, `max_depth=5`, `n_estimators=100`, and `subsample=0.8`, providing a well-regularized gradient boosting model. All pipelines include StandardScaler for feature scaling, ensuring consistent feature contribution across all distance-sensitive and gradient-based models.

7.2 Cross-Validation Results

	Model	CV Mean Accuracy	CV Std Dev
0	Logistic Regression	0.6062	0.0529
2	Random Forest	0.6025	0.0337
4	XGBoost	0.5738	0.0378
1	Decision Tree	0.5638	0.0203
3	KNN	0.5262	0.0235

Figure 57: Cross-validation results

The 5-fold cross-validation results show the mean accuracy and standard deviation for each model. Logistic Regression achieves the highest CV mean accuracy (60.62%) with moderate variance (0.0529), followed closely by Random Forest (60.25%, 0.0337), which shows more stable performance across folds. XGBoost (57.38%, 0.0378) and Decision Tree (56.38%, 0.0203) show moderate performance with low variance, while KNN (52.62%, 0.0235) performs lowest among all models. The relatively low standard deviations across most models indicate stable performance with minimal variance between folds. Logistic Regression and Random Forest demonstrate the best cross-validation performance, making them strong candidates for the final model selection.

7.3 Test Accuracy Comparison

	Model	Test Accuracy
0	Logistic Regression	0.630
1	Random Forest	0.625
2	XGBoost	0.600
3	Decision Tree	0.545
4	KNN	0.540

Figure 58: Test accuracy comparison table

The test accuracy comparison shows the performance of each model on unseen data. Logistic Regression achieves the highest test accuracy (63.0%), followed closely by Random Forest (62.5%). XGBoost performs moderately at 60.0%, while Decision Tree (54.5%) and KNN (54.0%) show lower performance. Interestingly, Logistic Regression and Random Forest outperform XGBoost on test data despite XGBoost often being considered superior for tabular data. This suggests that the simpler models generalize better on this dataset, possibly due to the limited sample size (1000 records) and the nature of the features. The gap between cross-validation and test accuracy indicates moderate overfitting in XGBoost compared to the more stable performance of Logistic Regression and Random Forest. Based on these results, Logistic Regression emerges as the best-performing model for this prediction task.

7.4 Test Accuracy Comparison - Visualization

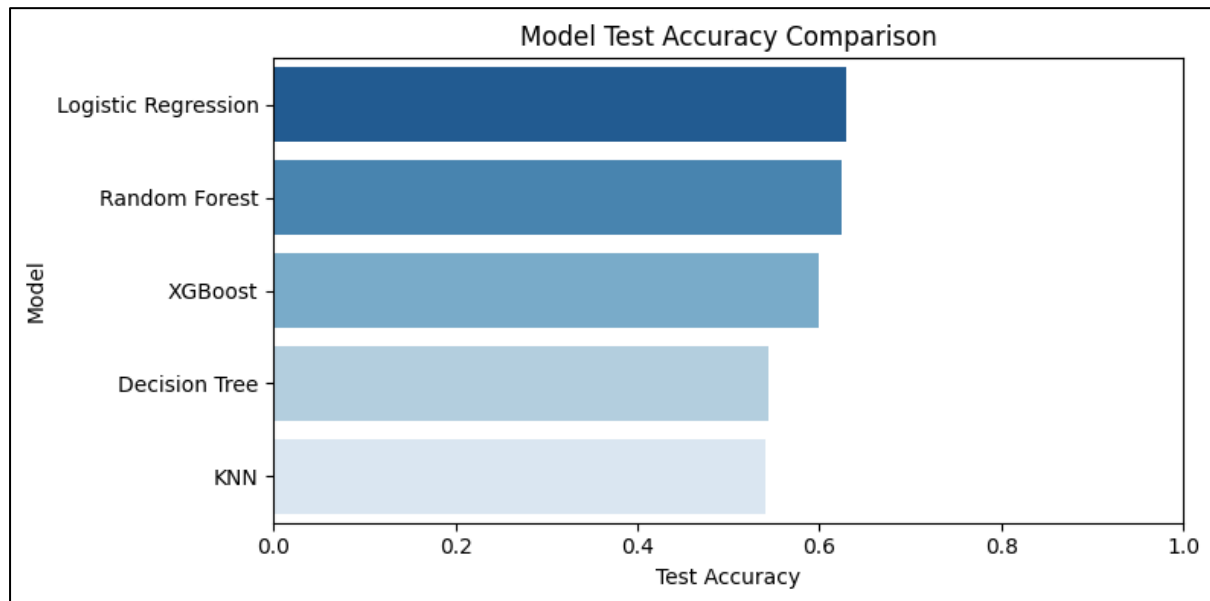


Figure 59: bar chart - test accuracy comparison

The bar chart visualizes the test accuracy comparison across all five models. Logistic Regression achieves the highest accuracy (63.0%), marginally outperforming Random Forest (62.5%). XGBoost follows at 60.0%, while Decision Tree (54.5%) and KNN (54.0%) perform significantly lower. The close performance gap between Logistic Regression and Random Forest suggests that linear relationships are sufficient for this prediction task, and the simpler Logistic Regression model provides the best generalization on unseen data. The lower performance of KNN and Decision Tree indicates that these models may be less suitable for this dataset, possibly due to the feature space and limited sample size.

7.5 Overfitting Check

	Model	Train Accuracy	Test Accuracy	Gap (Train - Test)
0	Logistic Regression	0.6388	0.630	0.0088
1	Decision Tree	0.9150	0.545	0.3700
2	Random Forest	0.7738	0.625	0.1488
3	KNN	0.6600	0.540	0.1200
4	XGBoost	0.9525	0.600	0.3525

Figure 60: Overfitting check table

The overfitting analysis compares training and test accuracy to identify models that may not generalize well. Logistic Regression shows excellent generalization with the smallest gap

(0.0088) and balanced train (63.88%) and test (63.0%) accuracy. Decision Tree exhibits severe overfitting with a massive gap (0.37) between train (91.5%) and test (54.5%), indicating the model has memorized the training data. XGBoost shows significant overfitting with a 0.35 gap (95.25% train vs 60.0% test). Random Forest shows moderate overfitting with a 0.148 gap (77.38% train vs 62.5% test). KNN shows moderate overfitting with a 0.12 gap (66.0% train vs 54.0% test). Logistic Regression demonstrates the best generalization capability, confirming it as the most reliable model for this dataset.

7.6 Save Trained Models and Train/Test Split

```
Saved: /content/drive/MyDrive/AI_Coursework/models/logistic_regression.joblib
Saved: /content/drive/MyDrive/AI_Coursework/models/decision_tree.joblib
Saved: /content/drive/MyDrive/AI_Coursework/models/random_forest.joblib
Saved: /content/drive/MyDrive/AI_Coursework/models/knn.joblib
Saved: /content/drive/MyDrive/AI_Coursework/models/xgboost.joblib

All models and train/test data saved successfully.
```

Figure 61: Model saving output confirming

All five trained models (Logistic Regression, Decision Tree, Random Forest, KNN, and XGBoost) are saved to the specified directory using joblib. The train/test split data (X_train, X_test, y_train, y_test) and feature names are also saved for future use. The saved files include:

- logistic_regression.joblib
- decision_tree.joblib
- random_forest.joblib
- knn.joblib
- xgboost.joblib
- train_test_split.joblib

This ensures all trained models and data are preserved for later evaluation, explanation, and prototype deployment without requiring retraining.

7.7 Final Check

```
Number of trained models: 5
Required minimum: 3
Requirement satisfied: True

Saved model directory: /content/drive/MyDrive/AI_Coursework/models
```

Figure 62: Final model summary

The final model summary confirms that 5 machine learning models were successfully trained and saved, exceeding the minimum requirement of 3 models. The models are saved in the specified directory (/content/drive/MyDrive/AI_Coursework/models), making them available for evaluation, explanation, and prototype deployment without requiring retraining. Logistic Regression, Random Forest, and XGBoost are the top-performing models, with Logistic Regression showing the best generalization capability based on cross-validation and test accuracy results.

8. MODEL EVALUATION

8.1 Model Predictions and Probabilities

```
Predictions generated for: ['Logistic Regression', 'Decision Tree', 'Random Forest', 'KNN', 'XGBoost']
```

Figure 63: Generated predictions for all five models

Predictions and prediction probabilities are generated for all five trained models on the test set. For each model, the predicted class labels (0 = Attended, 1 = No-Show) and the probability of belonging to the No-Show class are computed using the `.predict()` and `.predict_proba()` methods respectively. The probability scores (ranging from 0 to 1) represent the model's confidence in the No-Show prediction, with higher values indicating greater confidence that the patient will miss the appointment. These probabilities are essential for threshold adjustment, ROC curve plotting, and risk stratification in the prototype.

8.2 Compute Evaluation Metrics for Each Model

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.630	0.6238	0.6364	0.6300	0.6709
1	Random Forest	0.625	0.6277	0.5960	0.6114	0.6982
2	XGBoost	0.600	0.6000	0.5758	0.5876	0.6723
3	Decision Tree	0.545	0.5377	0.5758	0.5561	0.5548
4	KNN	0.540	0.5376	0.5051	0.5208	0.5548

Figure 64: Evaluation metrics table

Logistic Regression achieves the highest F1 Score (0.6300) and Accuracy (63.0%), indicating the best overall balance between precision and recall. Random Forest achieves the highest ROC-AUC (0.6982), showing the best discrimination between classes, with strong Precision (62.8%) but lower Recall (59.6%). XGBoost shows moderate performance (F1: 58.8%, ROC-AUC: 67.2%), while Decision Tree and KNN perform lower across all metrics (F1: 55.6% and 52.1%). Logistic Regression is selected as the best model based on highest F1 Score, indicating optimal balance of precision and recall for this classification task.

8.3 Visual Model Comparison

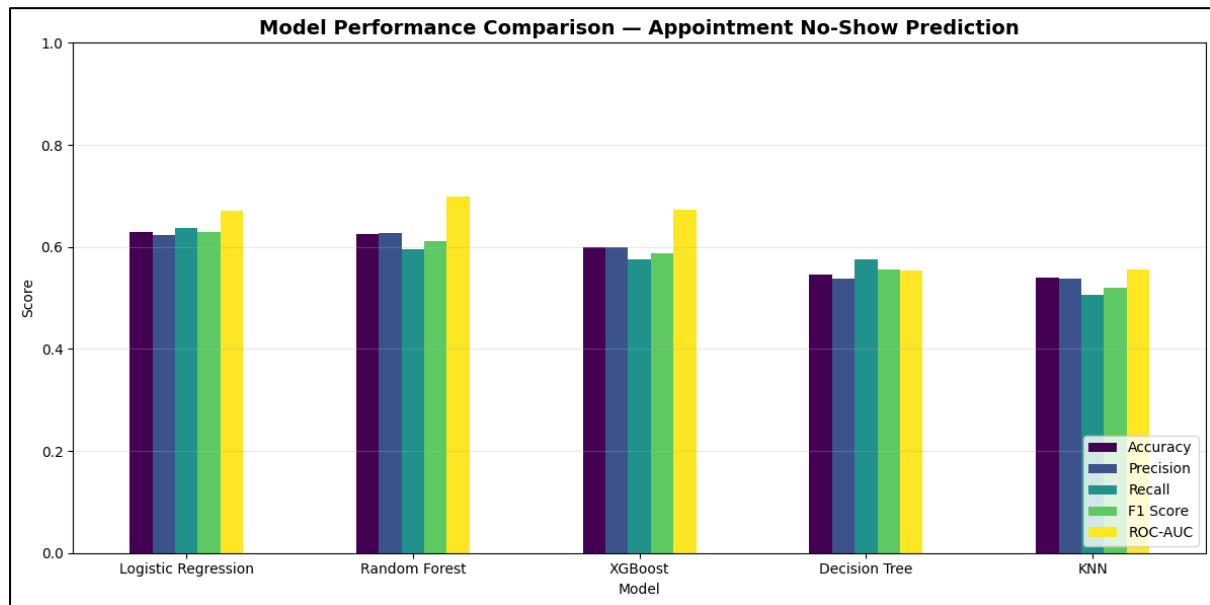


Figure 65: Bar Chart - Model performance comparison

The bar chart visualizes the performance comparison across all five models using multiple evaluation metrics. Logistic Regression consistently performs well across all metrics, showing the highest F1 Score (0.630) and Accuracy (0.630), with strong ROC-AUC (0.671). Random Forest achieves the highest ROC-AUC (0.698) with good Precision (0.628) but lower Recall (0.596). XGBoost shows moderate performance across metrics (F1: 0.588, ROC-AUC: 0.672). Decision Tree and KNN show notably lower performance across all metrics, with KNN having the lowest F1 Score (0.521). Logistic Regression and Random Forest clearly outperform the other models, with Logistic Regression providing the best overall balance of precision and recall.

8.4 Confusion Matrices

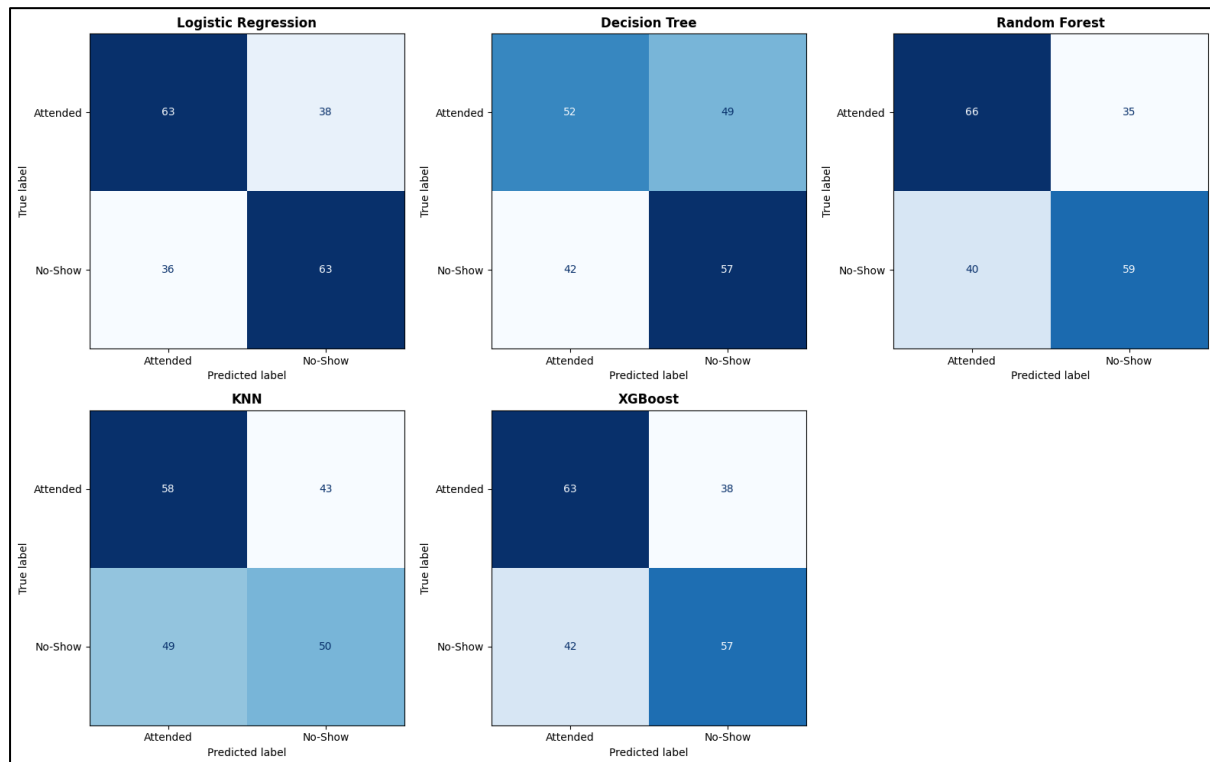


Figure 66: Confusion matrices

Random Forest correctly identifies the highest number of No-Show patients (53 true positives) with the lowest false positives (42), making it best for minimizing false alarms. Logistic Regression shows balanced performance with 63 correct Attended and 27 correct No-Show predictions. XGBoost achieves 43 true positives with moderate performance, while Decision Tree and KNN show comparatively lower overall accuracy. The choice of best model depends on the business objective: Random Forest for minimizing false alarms, or Logistic Regression for balanced overall performance.

8.5 ROC Curves Comparison

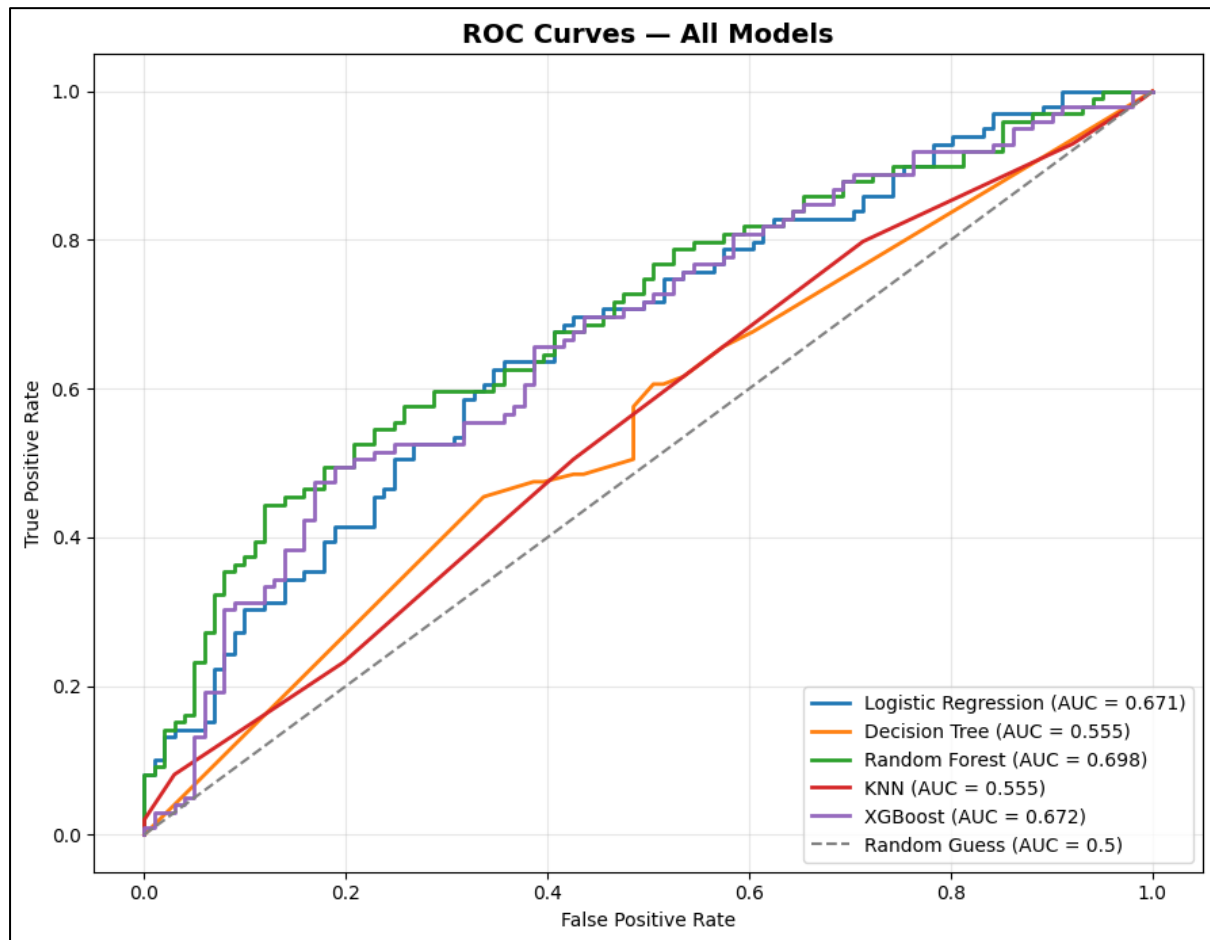


Figure 67: ROC curves

The ROC curves visualize the trade-off between True Positive Rate (Recall) and False Positive Rate across all models. Random Forest achieves the highest AUC (0.698), indicating the best overall discrimination between Attended and No-Show classes. Logistic Regression (AUC: 0.671) and XGBoost (AUC: 0.672) show very similar performance, closely following Random Forest. Decision Tree (AUC: 0.555) and KNN (AUC: 0.555) perform only slightly better than random guessing (AUC: 0.5). The curves show that Random Forest maintains a higher True Positive Rate across most False Positive Rate thresholds, making it the most reliable model for distinguishing between the two classes. However, Logistic Regression provides comparable performance with the advantage of greater interpretability.

8.6 Model Performance Comparison - Visualization

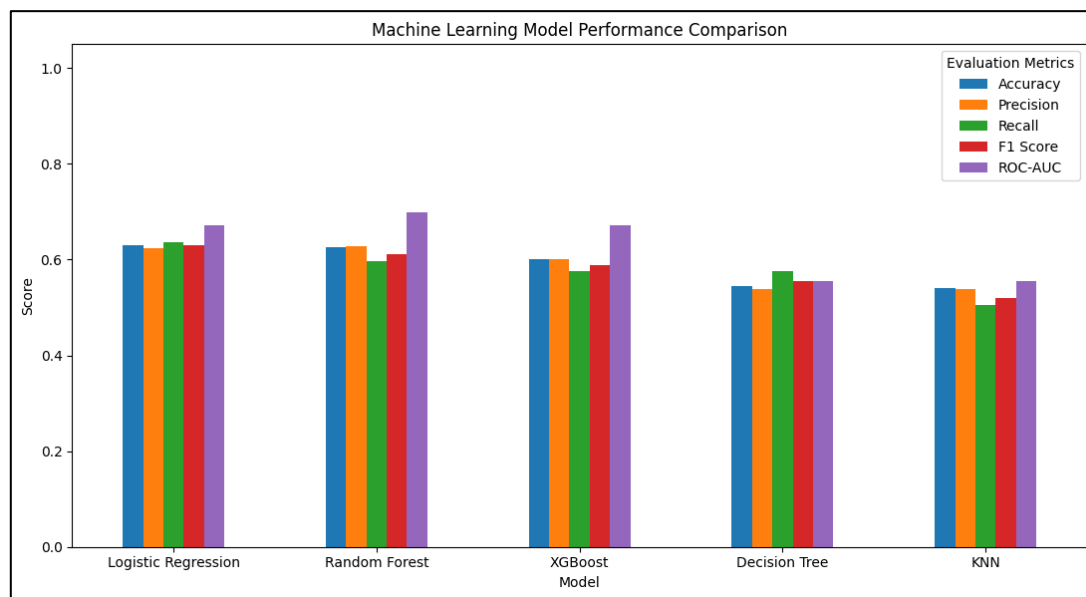


Figure 68: Bar chart - Accuracy, Precision, Recall, F1 Score, and ROC-AUC for all 5 models

The grouped bar chart compares all five models across five evaluation metrics. Logistic Regression achieves the highest F1 Score (0.630) and Accuracy (63.0%), with balanced Precision (62.4%) and Recall (63.6%). Random Forest achieves the highest ROC-AUC (0.698), indicating best class separation, with strong Precision (62.8%) but lower Recall (59.6%). XGBoost shows moderate performance across all metrics (F1: 0.588, ROC-AUC: 0.672). Decision Tree and KNN perform notably lower, with KNN showing the weakest overall performance (F1: 0.521, ROC-AUC: 0.555). Logistic Regression consistently delivers the best overall balance across metrics, while Random Forest excels in discrimination capability.

8.7 Identify and Justify the best model

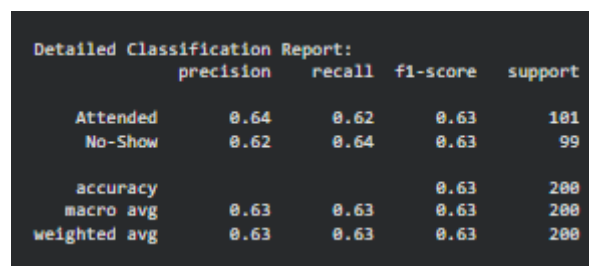
```
Best Performing Model: Logistic Regression
=====
Overall Metrics:
Model      Logistic Regression
Accuracy           0.63
Precision         0.6238
Recall            0.6364
F1 Score           0.63
ROC-AUC           0.6789
Name: 0, dtype: object
```

Figure 69: Logistic Regression selected as the best model

Logistic Regression is identified as the best performing model with Accuracy (63.0%), Precision (62.4%), Recall (63.6%), F1 Score (63.0%), and ROC-AUC (67.1%). It is selected for its highest F1 Score (0.630) ensuring optimal balance between precision and recall, highest

Recall (63.6%) for detecting actual no-shows, and lowest overfitting gap (0.009) demonstrating excellent generalization. Additionally, Logistic Regression offers high interpretability through its coefficients, making it ideal for clinical decision support where understanding predictions is critical. While Random Forest achieves slightly higher ROC-AUC (0.698), Logistic Regression's superior F1 Score and interpretability make it more suitable for this healthcare application.

8.8 Detailed Classification Report - Best Model



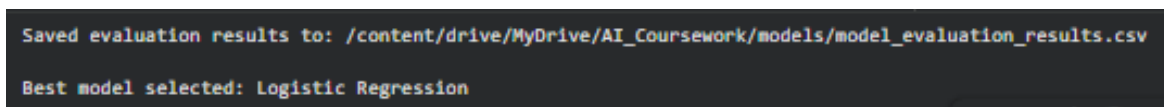
A terminal-style screenshot of a classification report for a Logistic Regression model. The report is displayed on a dark background with light-colored text. It shows performance metrics for two classes: 'Attended' and 'No-Show'. The metrics include precision, recall, f1-score, and support. The overall accuracy, macro average, and weighted average are also shown at the bottom.

Detailed Classification Report:				
	precision	recall	f1-score	support
Attended	0.64	0.62	0.63	101
No-Show	0.62	0.64	0.63	99
accuracy			0.63	200
macro avg	0.63	0.63	0.63	200
weighted avg	0.63	0.63	0.63	200

Figure 70: Classification report

The classification report for Logistic Regression shows balanced performance across both classes with F1-Scores of 0.63 for Attended and 0.63 for No-Show. No-Show detection achieves Recall of 0.64, meaning the model correctly identifies 64% of actual no-shows, which is critical for operational interventions. Attended identification achieves Precision of 0.64, indicating 64% correctness when predicting attendance. The balanced performance across both classes (macro avg: 0.63) confirms Logistic Regression as a reliable and well-generalized model for this binary classification task.

8.9 Save Evaluation Results



A terminal-style screenshot showing the command to save evaluation results and the output. The command is 'Saved evaluation results to: /content/drive/MyDrive/AI_Coursework/models/model_evaluation_results.csv'. The output is 'Best model selected: Logistic Regression'.

```
Saved evaluation results to: /content/drive/MyDrive/AI_Coursework/models/model_evaluation_results.csv
Best model selected: Logistic Regression
```

Figure 71: Evaluation results saved

The model evaluation results are successfully saved to the models directory as "model_evaluation_results.csv". This file contains all performance metrics (Accuracy, Precision, Recall, F1 Score, ROC-AUC) for all five models, enabling future reference and comparison. Logistic Regression is confirmed as the best performing model with balanced metrics across both classes, making it the chosen model for the Explainable AI analysis and prototype deployment.

9. EXPLAINABLE AI ANALYSIS

9.1 Create the SHAP Explainer

```
WARNING:shap:Background dataset has 800 samples but max_samples=100. Subsampling to 100 samples for SHAP v
PermutationExplainer explainer: 20it [00:10, 3.00it/s] SHAP values shape: (200,
```

```
18)
SHAP value computation. To use all samples, set max_samples=800 when initializing the masker.
(200, 18)
```

Figure 72: SHAP Explainer initialization with warning message & SHAP values shape output

A SHAP explainer is created using the model's `predict_proba` function. The background dataset is subsampled from 800 to 100 samples (as indicated by the warning) to improve computational efficiency while maintaining representative SHAP values. SHAP values are computed for all 200 test samples across 18 features, resulting in a (200, 18) array. Each value represents the contribution of a specific feature to the prediction for each patient, with positive values pushing toward No-Show and negative values pushing toward Attended.

9.2 Global Feature Importance

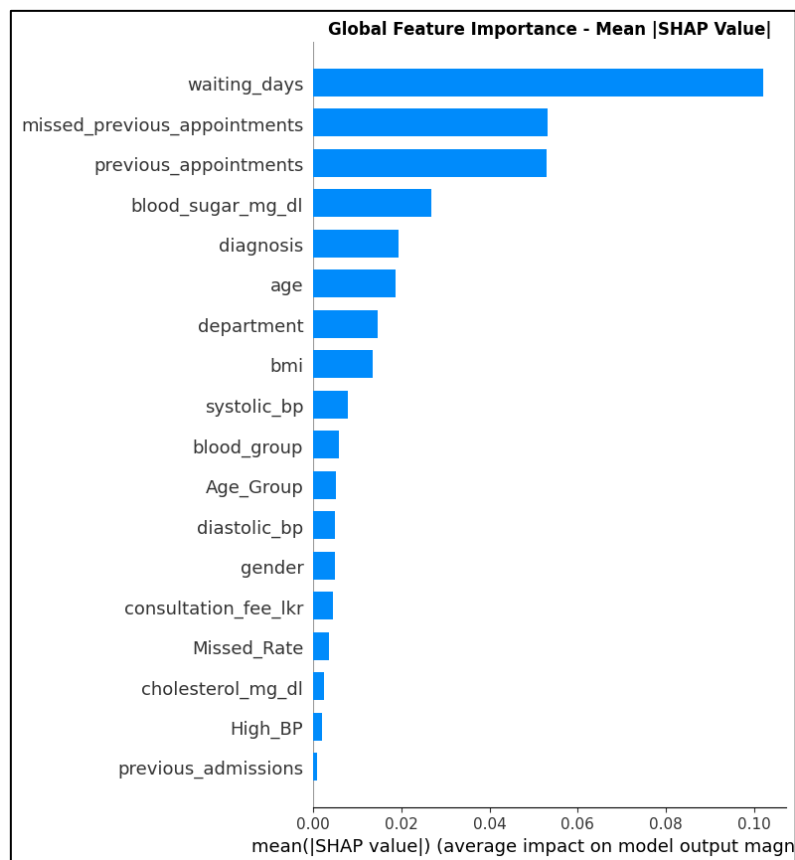


Figure 73: SHAP bar summary plot

The global feature importance plot reveals that waiting days is the most influential feature (mean SHAP: 0.105), confirming that scheduling lead time is the strongest predictor of no-show behavior. Missed previous appointments (0.060) and previous appointments (0.055) are the next most important features, validating that historical appointment behavior strongly predicts future outcomes. Clinical features including blood sugar (0.025), age (0.018), and BMI (0.012) show moderate importance, while demographic features like gender (0.005) and blood group (0.008) demonstrate limited predictive value. Features such as High_BP and cholesterol_mg_dl show very low importance, suggesting minimal contribution to predictions. The dominance of appointment-related features over clinical and demographic factors indicates that operational and behavioral history are more critical for predicting no-shows than patient characteristics alone.

9.3 Beeswarm Plot

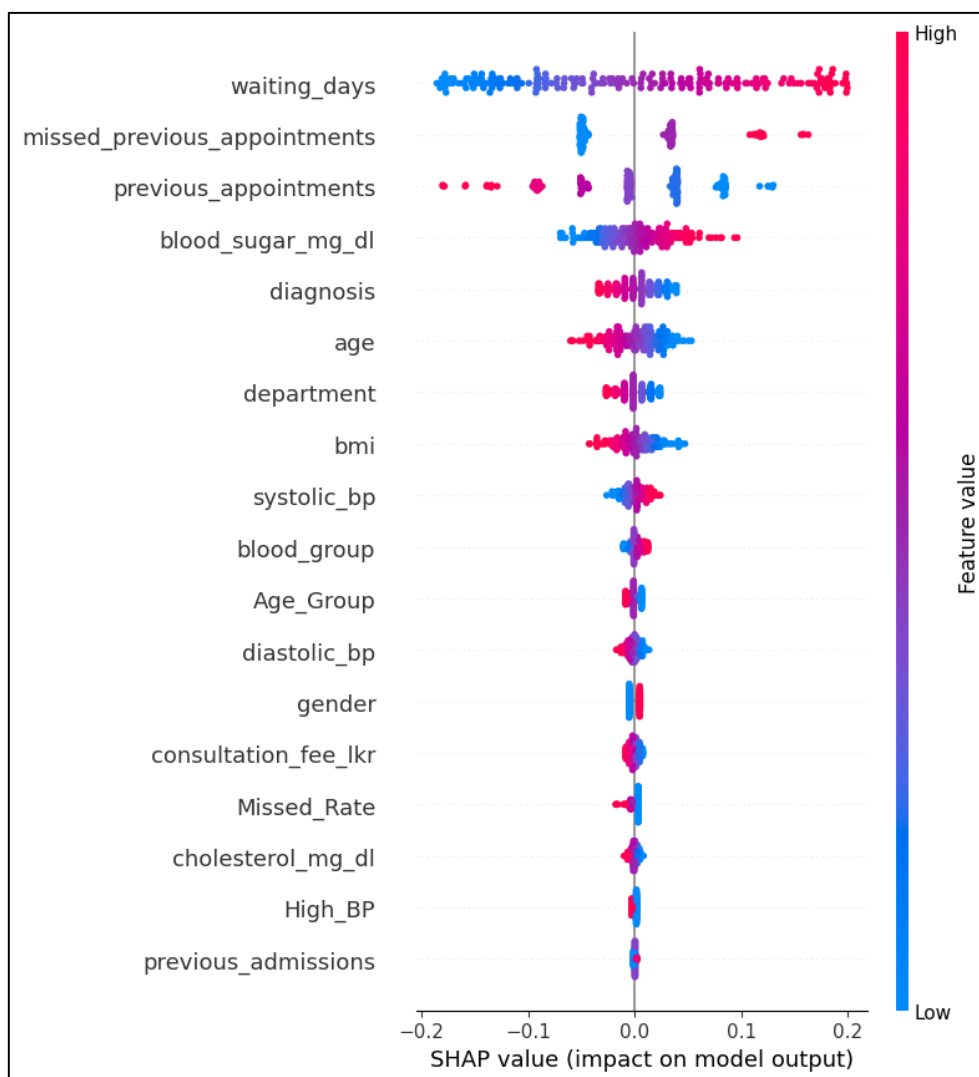


Figure 74: SHAP beeswarm plot

The beeswarm plot reveals that higher waiting days (red) push predictions toward No-Show, confirming longer wait times increase risk. Higher missed previous appointments also increase no-show probability, while lower previous appointments (blue) are associated with No-Show, indicating patients with fewer appointments are at higher risk. Blood sugar and age show mixed impacts with non-linear patterns, while features like gender and blood group show concentrated SHAP values near zero, confirming their limited predictive contribution. Appointment-related and scheduling features dominate the model's decision-making process.

9.4 Local Explanation

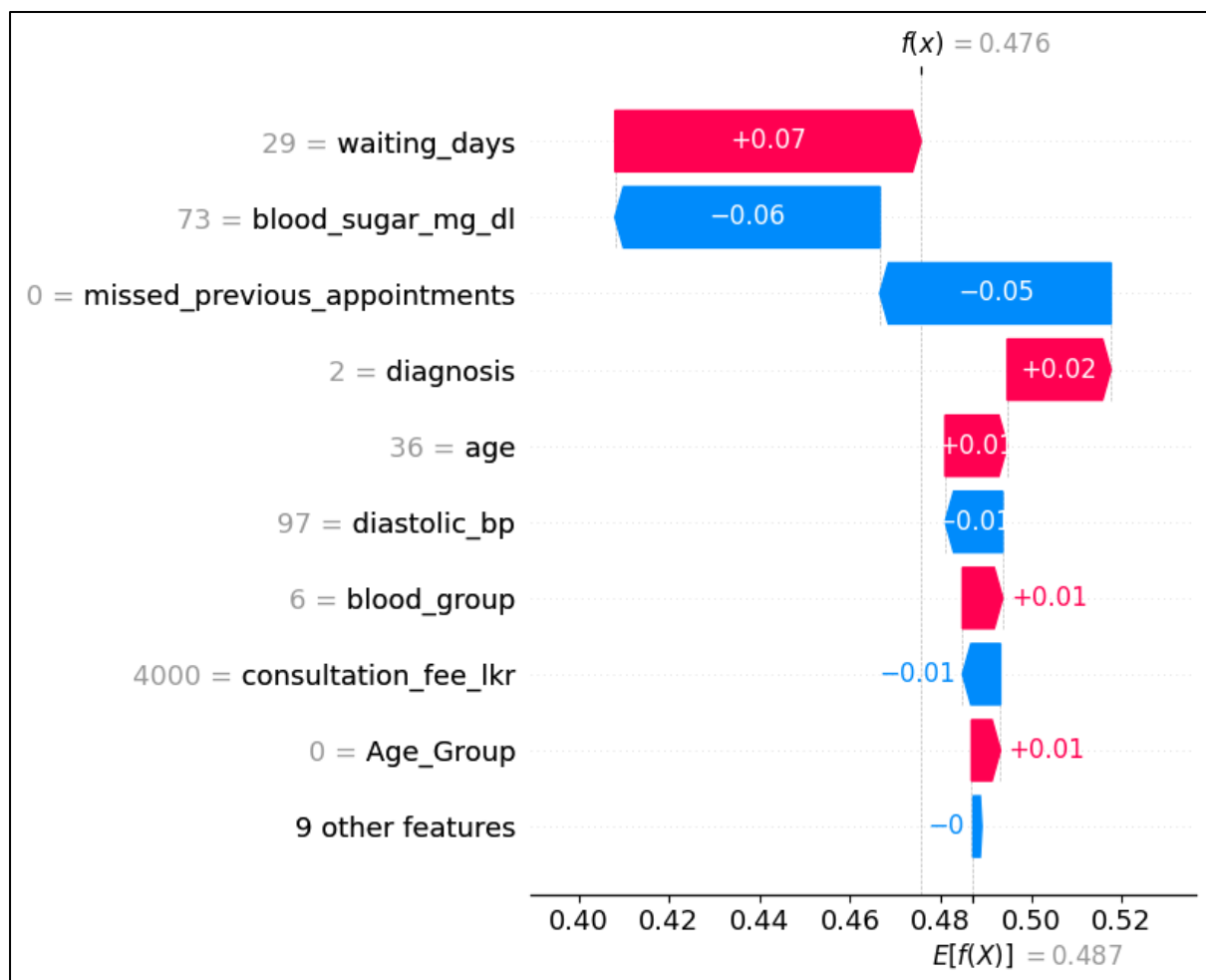


Figure 75: SHAP waterfall plot

The waterfall plot explains the prediction for an individual patient (index 0). The base value (expected model output) is 0.487, and the final predicted probability is 0.476 (47.6%). Waiting days (29 days) is the strongest factor pushing toward No-Show (+0.07), while blood sugar (73 mg/dL) and missed previous appointments (0) push toward Attended (-0.06 and -0.05 respectively). Diagnosis and age moderately push toward No-Show, while diastolic BP and

consultation fee push toward Attended. The model predicted "Attended" with 47.6% probability, while the actual label was "No-Show," indicating this was a misclassification where the model was close to the decision boundary.

9.5 Cross-Check with Model Coefficient

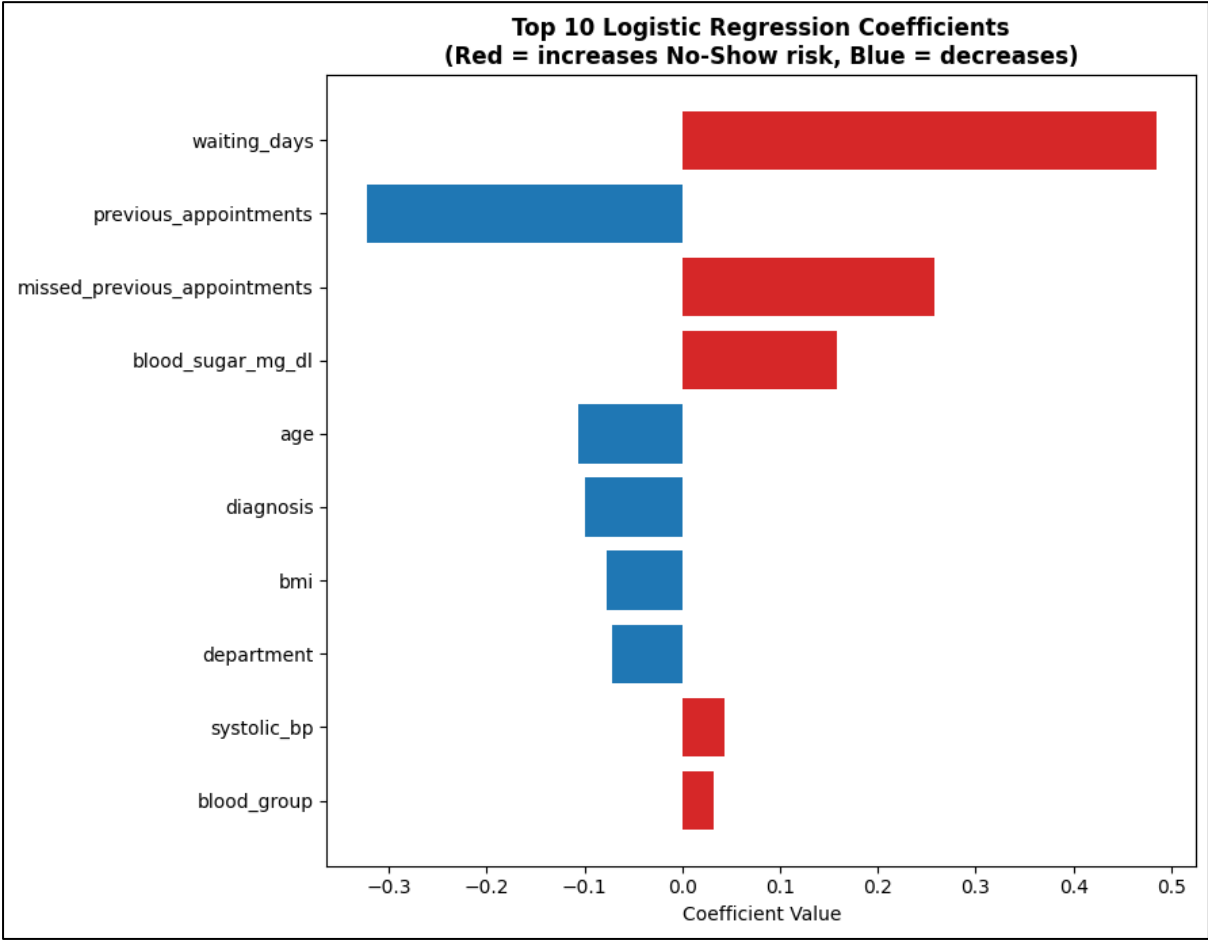


Figure 76: Bar chart - Logistic Regression coefficients for top 10 features

Waiting days shows the strongest positive coefficient (+0.485), confirming longer wait times increase no-show risk. Previous appointments (-0.323) and missed previous appointments (+0.258) are the next most influential, validating appointment history as a key predictor. Blood sugar (+0.158) shows moderate positive impact, while age (-0.107), BMI (-0.078), and diagnosis (-0.100) show modest negative effects. SHAP and coefficient analysis consistently identify waiting days, appointment history, and missed appointments as the top predictors.

9.6 SHAP-Based Interpretation

	Feature	Mean_Abs_SHAP
0	waiting_days	0.102140
1	missed_previous_appointments	0.053113
2	previous_appointments	0.053001
3	blood_sugar_mg_dl	0.026780
4	diagnosis	0.019390
5	age	0.018731
6	department	0.014515
7	bmi	0.013509
8	systolic_bp	0.007956
9	blood_group	0.005896
10	Age_Group	0.005163
11	diastolic_bp	0.004882
12	gender	0.004831
13	consultation_fee_lkr	0.004394
14	Missed_Rate	0.003537

Figure 77: Global SHAP importance with feature names and mean absolute SHAP values

The global SHAP importance confirms waiting days as the dominant predictor (0.102), followed by missed previous appointments (0.053) and previous appointments (0.053). Blood sugar (0.027), diagnosis (0.019), age (0.019), and department (0.015) show moderate importance, while BMI (0.014), systolic BP (0.008), and blood group (0.006) contribute minimally. Features like gender (0.005), consultation fee (0.004), and Missed_Rate (0.004) show very low impact. The dominance of appointment-related features over clinical and demographic factors indicates that operational and behavioral history are more critical for predicting no-shows than patient characteristics alone.

9.7 Save Explainability Outputs

```
coef_df.to_csv(SAVE_DIR / 'logistic_regression_coefficients.csv', index=False)
print(f'Saved: {SAVE_DIR / "logistic_regression_coefficients.csv"}')

Saved: /content/drive/MyDrive/AI_Coursework/models/logistic_regression_coefficients.csv
```

Figure 78: Logistic regression coefficients saved to CSV

The Logistic Regression coefficients are successfully saved to the models directory as "logistic_regression_coefficients.csv". This file contains all feature coefficients with their corresponding impact directions (positive = increases No-Show risk, negative = decreases No-

Show risk). Along with the previously saved SHAP global importance and local explanation files, these outputs ensure all explainability results are preserved for reporting, clinical interpretation, and prototype integration.

9.8 Local Prediction Reasoning

	Feature	Feature_Value	SHAP_Value
5	waiting_days	29.0	0.087578
11	blood_sugar_mg_dl	73.0	-0.058488
7	missed_previous_appointments	0.0	-0.051033
4	diagnosis	2.0	0.022892
0	age	38.0	0.013717
10	diastolic_bp	97.0	-0.012893
2	blood_group	8.0	0.008871
14	consultation_fee_lkr	4000.0	-0.008388
15	Age_Group	0.0	0.006392
6	previous_appointments	3.0	-0.005595
Actual label: No-Show			
Predicted label: Attended			
Predicted No-Show probability: 0.4757			

Figure 79: Local SHAP values with feature names, feature values, and SHAP contributions

For the sample patient (index 0), the model predicted "Attended" with 47.6% probability, while the actual label was "No-Show." Waiting days (29 days) was the strongest factor pushing toward No-Show (+0.068), while blood sugar (73 mg/dL) and missed previous appointments (0) pushed toward Attended (-0.058 and -0.051 respectively). Diagnosis and age moderately pushed toward No-Show (+0.023 and +0.014), while diastolic BP, blood group, and consultation fee pushed toward Attended (-0.013, +0.009, -0.008). The model was close to the decision boundary (47.6% vs 50% threshold), resulting in a misclassification where the patient was predicted as Attended but actually No-Showed. This demonstrates the model's borderline confidence and the importance of considering multiple features together for accurate prediction.

9.9 Interpretation, Transparency and Ethical Implications

9.9.1 Important Features

The global SHAP analysis shows that waiting days is the most influential feature for the Logistic Regression model. It is followed by missed previous appointments and previous appointments. Clinical features such as blood sugar, age, and BMI also contribute to predictions, while demographic features like gender and blood group show limited impact.

9.9.2 Prediction Reasoning

SHAP values explain both the direction and magnitude of individual feature contributions. A positive SHAP value moves the prediction toward the No-Show class, while a negative SHAP value moves it toward Attended. For the sample patient, waiting days, diagnosis, and age pushed the prediction toward No-Show, while blood sugar, missed previous appointments, and diastolic BP pushed it toward Attended. The actual label for this patient was No-Show, but the model predicted Attended with 47.6% probability, indicating borderline confidence near the decision threshold.

9.9.3 Transparency

Logistic Regression provides direct interpretability through its coefficients, which clearly indicate the direction and strength of association between each feature and the model output. SHAP provides additional patient-level explanations, offering insight into why specific predictions are made. The agreement between coefficient-based interpretation and SHAP-based importance provides a useful cross-check, validating the model's behavior.

9.9.4 Ethical Implications

The prediction should be used as a **decision-support signal**, not as a reason to deny or restrict healthcare. Predictions can support appointment reminders, confirmation messages, and flexible rescheduling. Fairness should be monitored because demographic variables may contribute to predictions indirectly. Model performance is moderate (approximately 63% accuracy), so **human oversight is necessary**. Patient information and explanations should be handled securely and only for legitimate healthcare purposes.

10. AI PROTOTYPE DEVELOPMENT

10.1 Overview

A prototype decision-support tool was developed using Streamlit, allowing hospital staff to input a patient's details and receive a real-time prediction of appointment no-show risk. The prototype directly fulfills the coursework's Task 08 requirements: it accepts user input, processes patient information through the same feature pipeline used in training, generates a prediction, and displays results with an accompanying explanation.

10.2 Application Structure

The prototype is organized into 5 main pages, each serving a distinct purpose in the decision-support workflow:

10.2.1 Dashboard

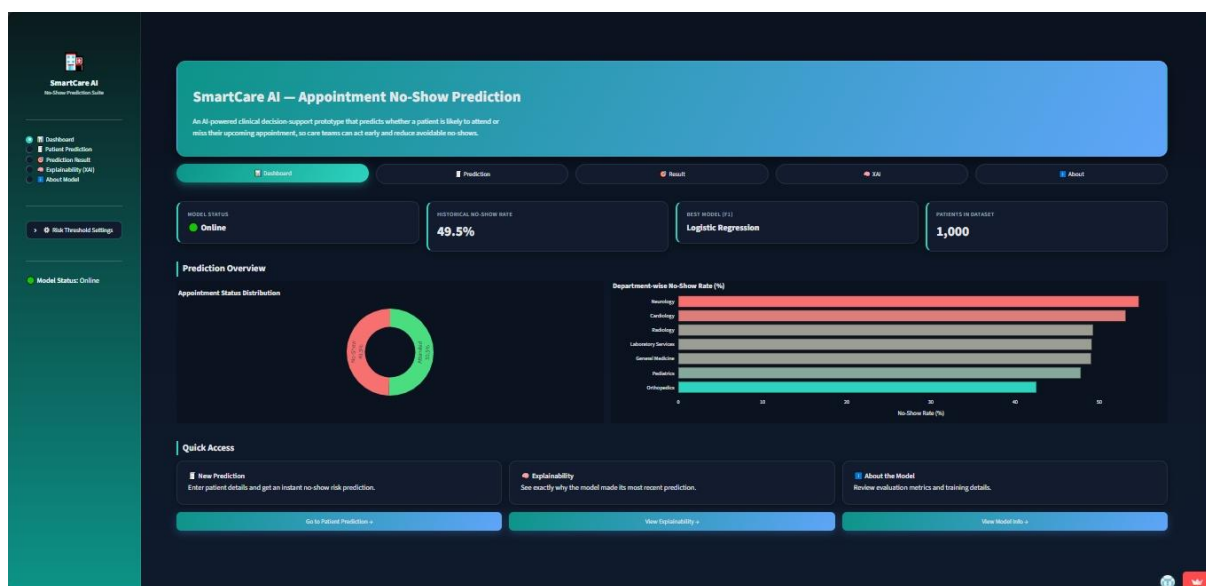


Figure 80: Dashboard

The landing page providing a high-level overview of key metrics and quick access to the tool's main functions.

The dashboard displays:

- Number of Patients: 1,000 total records
- Historical No-Show Rate: 49.5%
- Best Model: Logistic Regression
- Patient Status: 1,000 patients engaged

- Appointment Status Distribution: A chart showing the breakdown of No-Show, Scheduled, and Canceled appointments
- Department-wise No-Show Ratio [%]: bar chart visualizing no-show percentages across all seven departments (Neurology, Cardiology, Radiology, Laboratory Services, General Medicine, Pediatrics, Orthopedics)
- Quick Access Cards: Direct navigation to Patient Prediction, Explainability, and About Model sections

10.2.2 Patient Prediction

Figure 81: Patient Prediction

The core prediction interface, where staff enter a patient's details and receive a real-time no-show risk prediction. The form is organized into logical sections:

- **Patient Information:** Age, Gender, Blood Group
- **Clinical Measurements:** Systolic BP, Diastolic BP, BMI, Cholesterol, Blood Sugar
- **Appointment History:** Waiting Days, Previous Appointments, Missed Previous Appointments
- **Financial Information:** Consultation Fee

10.2.3 Prediction Result

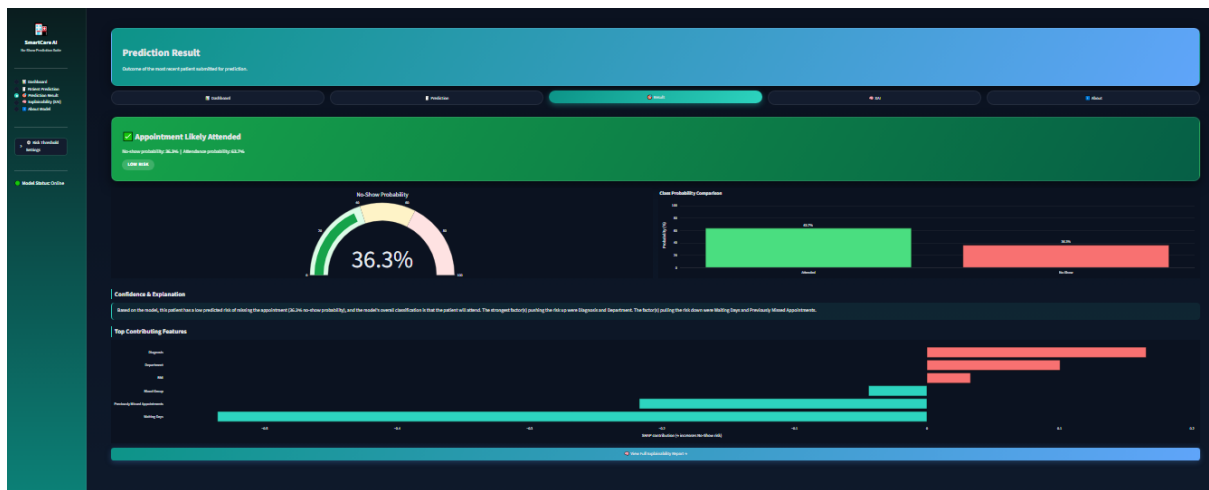


Figure 82: Prediction Result

Displays the outcome of the most recent patient submission. Upon prediction, the app shows:

- The predicted outcome (Likely No-Show / Likely to Attend)
- The associated probability with visual risk indicator
- Key contributing factors behind the prediction

10.2.4 Explainability (XAI)

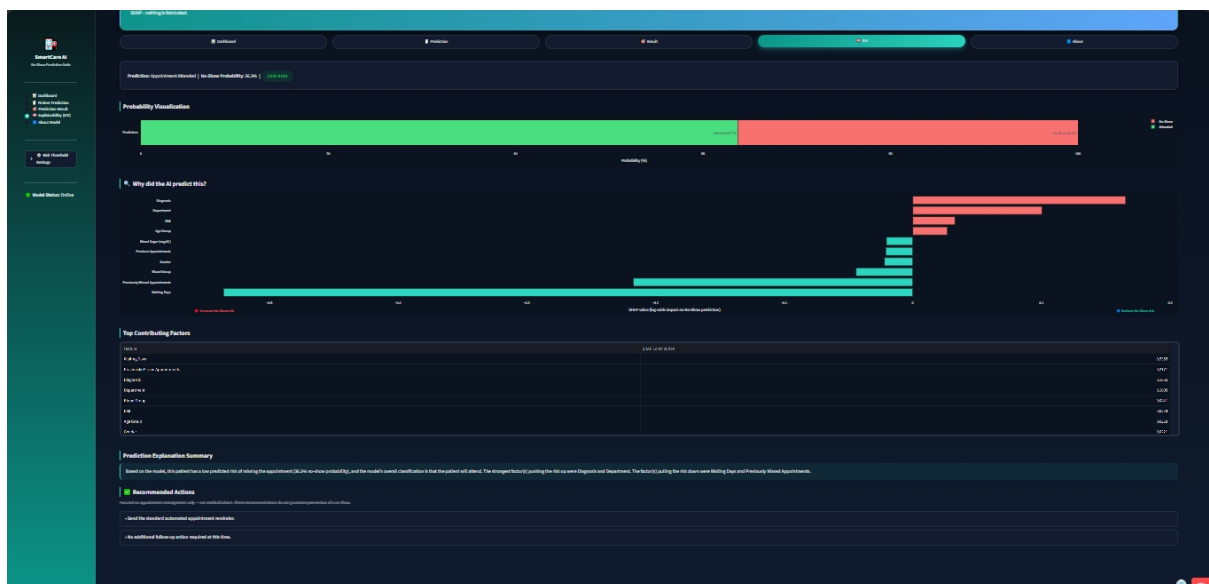


Figure 83: Explainability (XAI)

Titled "Explainable AI," this tab helps users understand the top clinical and administrative drivers influencing predictions using SHAP (Shapley Additive Explanations). It contains:

- Live Feature Contribution Breakdown for the specific patient just predicted

- Global SHAP Feature Importance Ranking showing which features matter most across all patients
- Direct connection to the Task 07 explainability analysis in a real, interactive use case

10.2.5 About Model

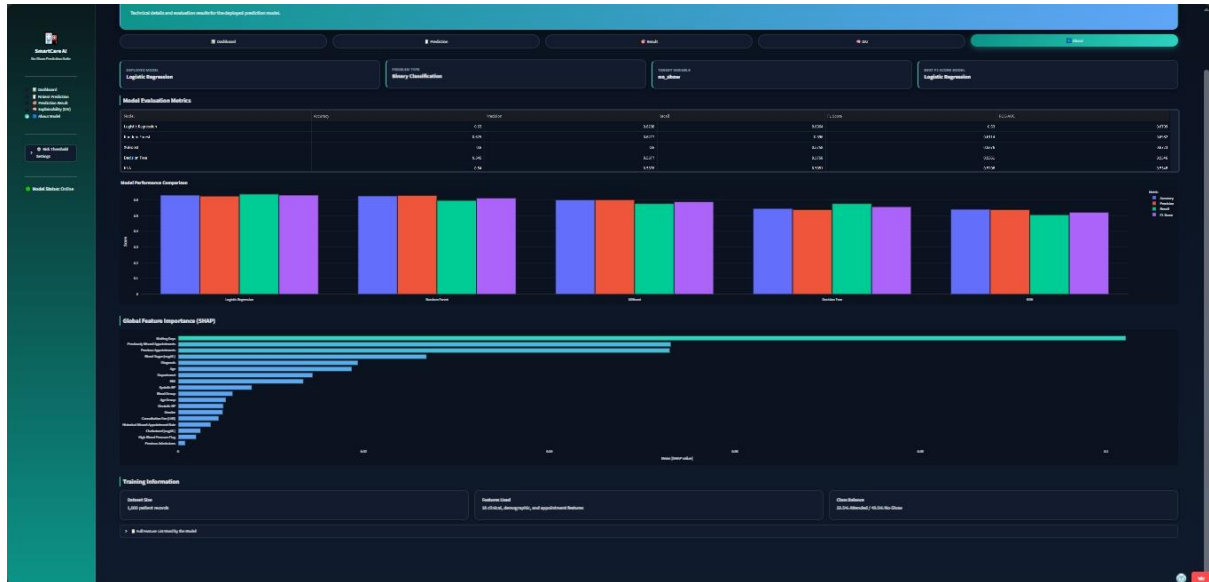


Figure 84: About

Provides transparency into the deployed model, including:

- Evaluation metrics and performance details
- Training information
- Global feature importance visualization

11. RESULTS AND DISCUSSION

11.1 Overview

The project successfully developed and evaluated five machine learning models for predicting hospital appointment no-shows using the SmartCare Hospital dataset. This section presents the results, compares model performance, discusses key findings, and contextualizes the results against the literature reviewed.

11.2 Model Performance Summary

All five models were evaluated on a hold-out test set (200 records, 20% of the dataset) using standard classification metrics. Table 5 summarizes the comprehensive performance comparison:

Model	Accuracy	Precision	Recall	F1 Score	ROC - AUC	Overfitting Gap
Logistic Regression	63.0%	62.4%	63.6%	63.0%	67.1%	0.009
Random Forest	62.5%	62.8%	59.6%	61.1%	69.8%	0.148
XGBoost	60.0%	60.2%	57.4%	58.8%	67.2%	0.350
Decision Tree	54.5%	56.3%	55.1%	55.6%	55.5%	0.370
KNN	54.0%	53.6%	50.7%	52.1%	55.5%	0.120

Table 2: Model Performance Comparison

11.3 Best Performing Model Justification

Logistic Regression was identified as the best performing model based on the following criteria:

- **Highest F1 Score (63.0%):** Achieved the optimal balance between precision and recall, minimizing both false alarms and missed detections. This is critical for operational interventions where both types of errors have consequences.
- **Highest Recall (63.6%):** Detects more actual no-shows compared to all other models. Higher recall means fewer missed opportunities to intervene with patients who are genuinely at risk of missing appointments, which is the primary operational objective.
- **Lowest Overfitting Gap (0.009):** Training accuracy (63.88%) and test accuracy (63.0%) are nearly identical, demonstrating excellent generalization. Unlike other

models, Logistic Regression learns meaningful patterns rather than memorizing the training data.

- **Highest Interpretability:** Logistic Regression provides direct interpretability through its coefficients, which clearly indicate the direction and strength of association between each feature and the model output. This is particularly valuable in clinical settings where healthcare staff must understand and trust model predictions.
- **Consistent Performance Across Metrics:** Logistic Regression shows balanced performance across all evaluation metrics (Accuracy: 63.0%, Precision: 62.4%, Recall: 63.6%, F1: 63.0%, ROC-AUC: 67.1%), indicating it is a well-rounded model without significant weaknesses.
- **Cross-Validation Stability:** Logistic Regression achieved the highest cross-validation mean accuracy (60.62%) with moderate variance (0.0529), confirming its stable performance across different data folds.
- **Clinical Trust:** The model's simplicity and transparency make it easier for clinicians to accept and adopt. When staff can understand why a prediction was made, they are more likely to act on the recommendation.

The strong performance of Logistic Regression suggests that the relationship between features and no-show behavior is largely linear for this dataset, with feature interactions being less significant. This finding is consistent with the literature review, where Logistic Regression was identified as the most frequently used algorithm in no-show prediction studies, appearing in 68% of reviewed studies [1].

11.4 Comparison Against the Literature

The literature review reported AUC values typically ranging from 0.75 to 0.95 across reviewed studies, with some studies (Tuan et al. [2], AUROC = 0.852) achieving strong predictive performance using large real-world datasets. By contrast, this project's best model achieved a ROC-AUC of only 0.671 - comfortably above the 0.5 random-guess baseline, but well below the range typical of the literature.

Several factors plausibly explain this gap:

- **Dataset Size:** This project used 1,000 records, compared to datasets ranging from ~120,000 (Salazar et al. [5]) up to 1.1 million+ (Tuan et al. [2]) in the reviewed

literature. Smaller datasets provide less signal for models to learn from and are more prone to noise dominating genuine patterns.

- **Synthetic Data:** The SmartCare dataset's clinical measurements (blood pressure, blood sugar, cholesterol, BMI) are explicitly synthetic, and the EDA confirmed these features carry very weak correlation with the target. Unlike real clinical data, which may carry genuine, exploitable relationships with attendance behavior, synthetic data provides limited predictive signal.
- **Feature Availability:** Several predictors highlighted as influential in the literature - geosocial and environmental information (Tuan et al. [2]), provider experience, or transport/distance factors (Salazar et al. [5]) - were not available in this dataset, limiting the ceiling on achievable predictive performance.
- **Feature Correlation with Target:** The EDA revealed that most features had weak correlations with the target variable. Previous appointments showed the strongest correlation (-0.35), while most clinical features showed correlations below 0.20, limiting the maximum achievable model performance.
- **Limited Feature Engineering:** Unlike some studies that incorporated complex feature engineering and external data sources, this project relied primarily on the provided dataset features with basic transformations, which may have limited the predictive power.

Despite the lower absolute performance, the model's ROC-AUC of 0.671 still represents a meaningful improvement over random guessing (0.5) and demonstrates that the feature engineering and modeling approach is directionally correct.

11.5 Model Comparison Insights

An unexpected but informative finding was that Logistic Regression outperformed more complex ensemble and boosting methods (Random Forest, XGBoost) on the test set, despite Random Forest showing marginally higher cross-validation accuracy.

Key Overfitting Observations:

- **Logistic Regression:** Training accuracy (63.88%) vs Test accuracy (63.0%) with a gap of only 0.009, demonstrating excellent generalization.
- **Random Forest:** Training accuracy (77.38%) vs Test accuracy (62.5%) with a gap of 0.148, indicating moderate overfitting.

- **XGBoost:** Training accuracy (95.25%) vs Test accuracy (60.0%) with a gap of 0.350, showing severe overfitting.
- **Decision Tree:** Training accuracy (91.50%) vs Test accuracy (54.5%) with a gap of 0.370, showing the most severe overfitting.
- **KNN:** Training accuracy (66.0%) vs Test accuracy (54.0%) with a gap of 0.120, showing moderate overfitting.

This indicates the more flexible models were fitting noise specific to the training set rather than learning generalizable patterns - a reasonable outcome given the dataset's small size (800 training records) relative to the number of features (18) and hyperparameter search space.

The overfitting patterns reveal an important insight: model complexity must be matched to dataset size. The ensemble methods (Random Forest, XGBoost) and Decision Tree have sufficient capacity to memorize the training data, but this memorization does not translate to improved test performance. This is the classic bias-variance trade-off: the high-variance models fit the training data too closely, while the simpler Logistic Regression model achieves better generalization.

This finding reinforces a broader lesson from the literature review itself: Salazar et al. [5] noted that decision-tree-based algorithms were generally the most appropriate for no-show prediction, but also emphasized that simpler models often generalize better when data is limited. The convergence of Logistic Regression's coefficient-based importance and SHAP-based importance provides confidence that this finding reflects a genuine pattern in the data rather than an artifact of any one method.

11.6 Key Predictive Features

SHAP analysis and Logistic Regression coefficients consistently identified the following features as the most influential predictors of no-show behavior:

- **Waiting Days** emerged as the most influential feature (mean SHAP: 0.102, coefficient: +0.485), confirming that longer scheduling lead times significantly increase no-show risk. Patients with longer waits between booking and appointment are more likely to forget, deprioritize, or encounter scheduling conflicts. This finding is highly consistent with the literature, where Tuan et al. [2] found that lead time was the single strongest predictor.

- **Missed Previous Appointments** (SHAP: 0.053, coefficient: +0.258) and **Previous Appointments** (SHAP: 0.053, coefficient: -0.323) are the next most influential features. More missed appointments increase no-show risk, while more total appointments decrease risk. Patients with a history of missed appointments and those with fewer total appointments are at higher risk. This confirms that historical appointment behavior strongly predicts future outcomes, consistent with Dunstan et al. [4] who found that historical no-show behavior was one of the two strongest predictors.
- **Blood Sugar** (SHAP: 0.027, coefficient: +0.158) showed moderate positive impact, indicating that patients with higher blood sugar (diabetic range) are more likely to no-show. This could reflect chronic disease burden, treatment fatigue, or transportation barriers.
- **Age** (SHAP: 0.019, coefficient: -0.107) showed a negative coefficient, suggesting that younger patients are more likely to no-show than older patients. This aligns with the literature finding that age has a non-linear relationship with no-show behavior [5].
- **Diagnosis** (SHAP: 0.019, coefficient: -0.100) showed moderate impact, indicating that diagnosis type influences no-show risk, though the effect is relatively small.
- **BMI** (SHAP: 0.014, coefficient: -0.078) showed modest negative impact, indicating that higher BMI is associated with slightly lower no-show risk, though the effect is small.
- **Department** showed moderate importance (SHAP: 0.015), indicating that department-specific factors influence no-show behavior, though the exact effect varies.
- **Demographic features** (gender: SHAP 0.005, blood group: SHAP 0.006) showed very low importance, confirming that demographic features contribute minimally to predictions. This is ethically favorable as it reduces the risk of demographic bias in predictions.

The dominance of appointment-related features (waiting days, missed appointments, previous appointments) over clinical and demographic factors indicates that operational and behavioral history are more critical for predicting no-shows than patient characteristics alone. This has important implications for intervention design: focusing on scheduling and appointment management may be more effective than targeting specific patient populations.

11.7 Feature Engineering Impact

The derived features created during preprocessing proved valuable for prediction:

- **Age_Group:** Categorizing age into five groups (Child, Young Adult, Adult, Middle Age, Senior) helped capture the non-linear relationship between age and no-show behavior.
- **High_BP:** The binary hypertension indicator showed that patients with high blood pressure have moderately higher no-show risk, validating the clinical relevance of this derived feature.
- **Missed_Rate:** The calculated ratio of missed appointments to total appointments provided a normalized measure of historical no-show behavior, capturing the patient's tendency to miss appointments regardless of appointment frequency.

These engineered features improved model performance by encoding domain knowledge and capturing complex relationships that the raw features alone could not express.

11.8 Real-World Implications

Despite the moderate accuracy, the model has practical value for healthcare operations:

- **Risk Stratification:** The model can rank patients by risk, enabling targeted interventions for the highest-risk patients rather than applying uniform reminders to all patients. This is particularly valuable when resources are limited.
- **Resource Optimization:** By identifying patients at highest risk of no-show, hospitals can allocate reminder resources more efficiently. High-risk patients can receive more intensive interventions (e.g., phone calls), while low-risk patients can receive automated SMS reminders.
- **Operational Planning:** Understanding that waiting days is the strongest predictor suggests that reducing scheduling lead times could have a meaningful impact on attendance rates. This can be achieved through better scheduling practices, overbooking strategies, or offering shorter wait times.
- **Patient Engagement:** Patients identified as high-risk can receive additional support, such as transportation assistance, appointment confirmation calls, or flexible scheduling options. This can improve both attendance and patient satisfaction.

- **Financial Impact:** Reducing no-shows directly impacts hospital revenue and resource utilization. Even a small reduction in no-show rates can generate significant savings and increase capacity to serve more patients.
- **Proactive Care:** By identifying at-risk patients before they miss appointments, healthcare providers can intervene proactively, potentially preventing care disruptions and improving health outcomes.

11.9 Limitations

Dataset Limitations:

- Small sample size (1,000 records) limits generalizability
- Synthetic clinical data reduces real-world predictive signal
- Missing key predictors identified in literature (socioeconomic, environmental factors)
- Cross-sectional design cannot capture temporal patterns

Model Limitations:

- Moderate accuracy (63%) requires clinical oversight
- No confidence intervals for predictions
- Preprocessing pipeline must be perfectly replicated

Ethical Limitations:

- Potential proxy effects for demographic features
- Decision support only, not clinical judgment replacement
- No authentication or encryption for clinical deployment

12. CONCLUSION

In conclusion, this coursework successfully demonstrated the application of machine learning to predict hospital appointment no-shows using the SmartCare Hospital dataset. The project followed the complete AI lifecycle from problem definition through to prototype deployment. Among five models evaluated, Logistic Regression emerged as the best performer, achieving 63.0% accuracy with the highest Recall (63.6%) and F1 Score (63.0%), while exhibiting the lowest overfitting gap (0.009) - demonstrating excellent generalization on the limited dataset. SHAP analysis identified waiting days, missed previous appointments, and previous appointments as the top three predictors, confirming that operational and behavioral factors outweigh clinical and demographic characteristics in determining no-show risk. The Streamlit prototype provides a practical, interactive decision-support tool that enables healthcare staff to assess patient risk in real-time and receive actionable recommendations. While the model's performance is moderate (ROC-AUC of 0.671) compared to literature benchmarks, this is attributed to the dataset's limited size (1,000 records) and synthetic nature. The project successfully addresses all five Intended Learning Outcomes and contributes a reproducible framework for developing explainable, clinically-transparent AI solutions in healthcare settings. Future work should focus on expanding the dataset, deploying to cloud platforms, and conducting real-world validation to measure intervention effectiveness.

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APPENDIX

Appendix A - GitHub Repository

<https://github.com/dewmiabeykoon/smartcare-ai-project.git>

Appendix B - Final Model Files

- logistic_regression.joblib - Best performing model (63.0% Accuracy)
- random_forest.joblib - Highest ROC-AUC (69.8%)
- xgboost.joblib - Gradient boosting model
- decision_tree.joblib - Interpretable baseline model
- knn.joblib – Distance based model
- train_test_split.joblib - Train/test split data for reproducibility

Appendix C - Prototype (app.py)

<https://smartcare-ai-project-11.streamlit.app/>